1) Muhammad Umer (22k-4160)
2) Siraj Ahmed (22k-4410)

BSCS-7C

Information Security Project / Assignment 4

Miss Yusra Kaleem

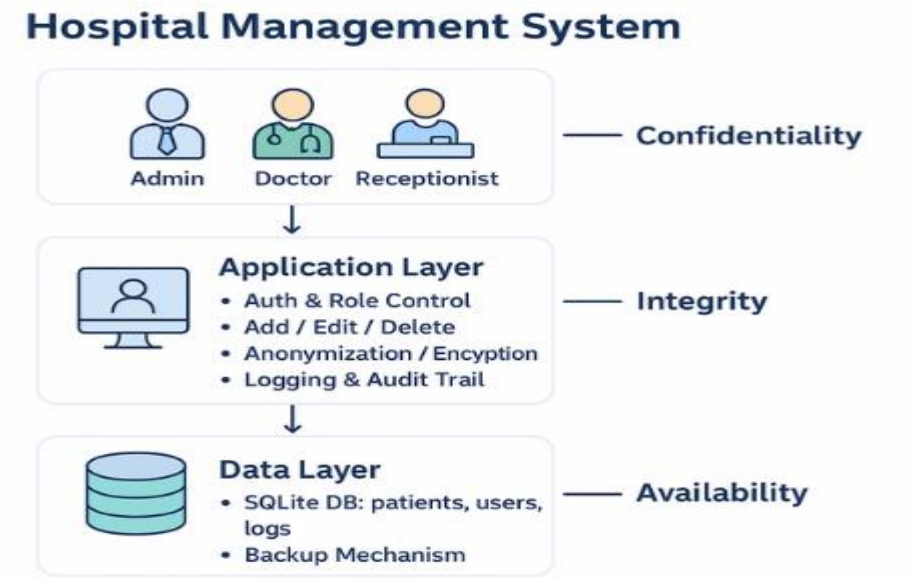
1. System Overview

The hospital management system is designed to transition from paper-based patient records to a digital platform, implementing privacy, security, and compliance measures aligned with GDPR and the CIA triad principles.

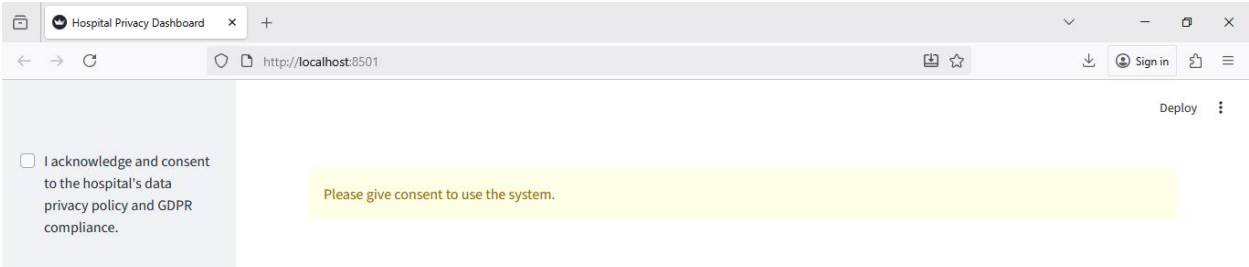
1.1 System Architecture Diagram (CIA Layers)

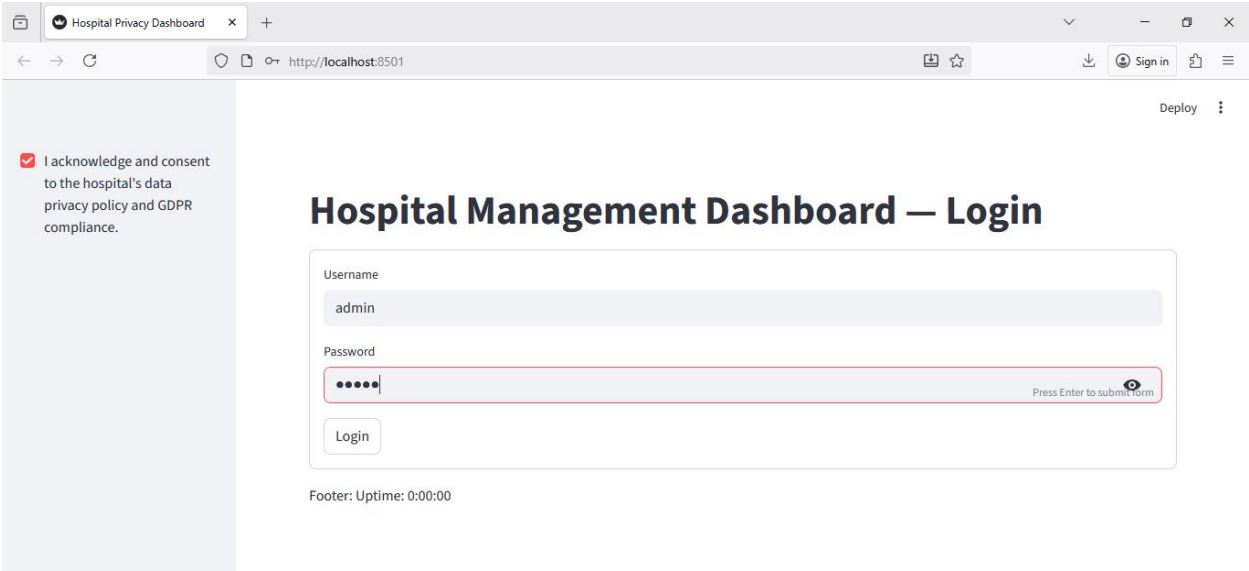
Description of the diagram:

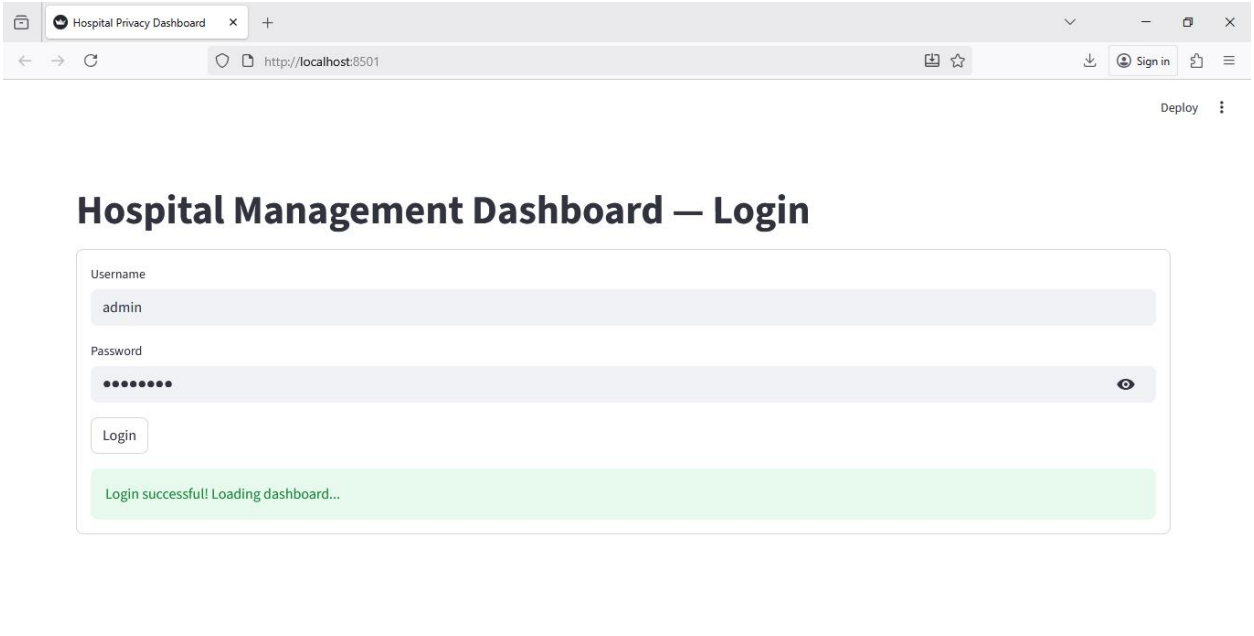
- **Confidentiality Layer:** Sensitive patient data (name, contact, diagnosis) is anonymized or encrypted. Role-based access ensures only authorized users can view data.
- **Integrity Layer:** All actions (add, edit, delete, anonymize, export) are logged with user ID, role, and timestamp to maintain data integrity.
- **Availability Layer:** System uptime is tracked, and database backups ensure data availability in case of failure.



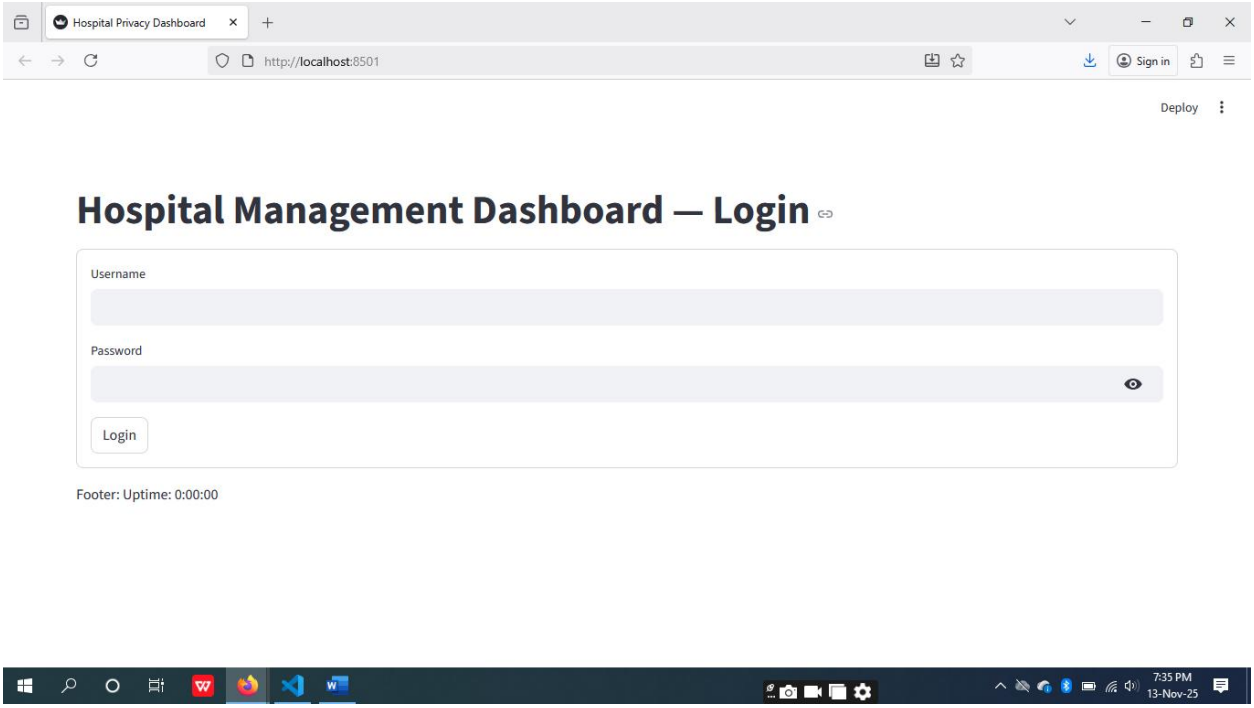
2. Screenshots of Key Functionalities

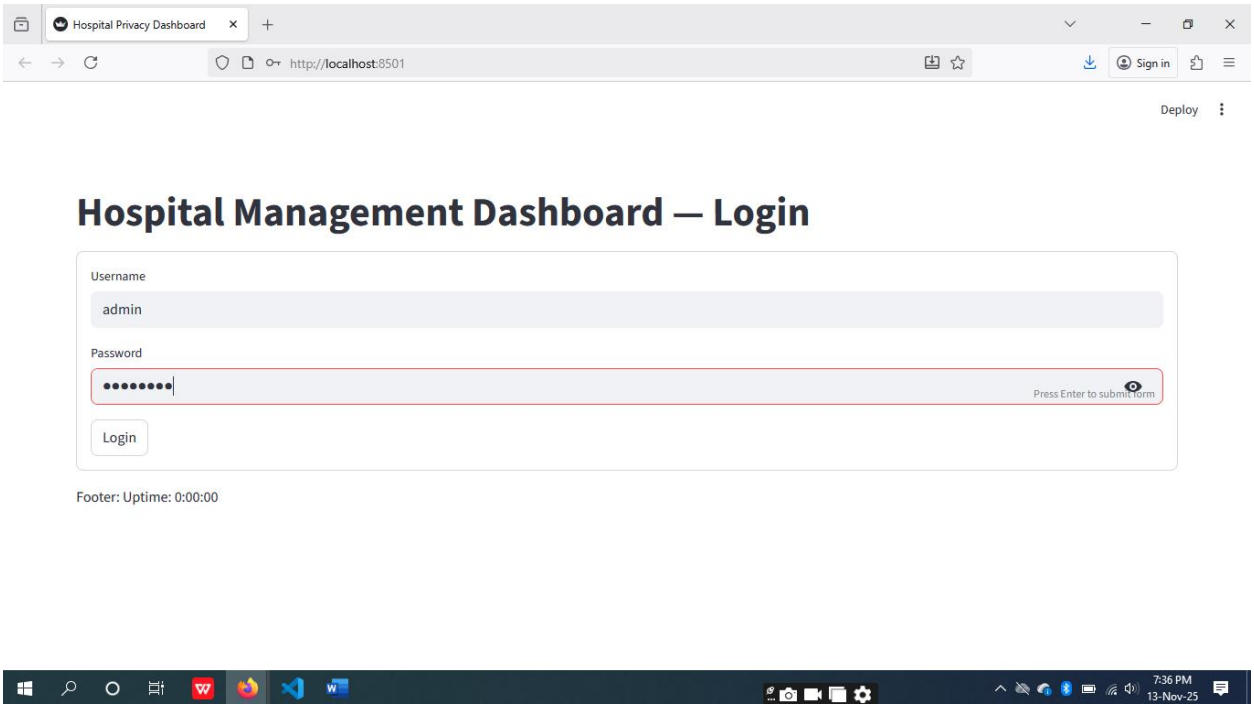


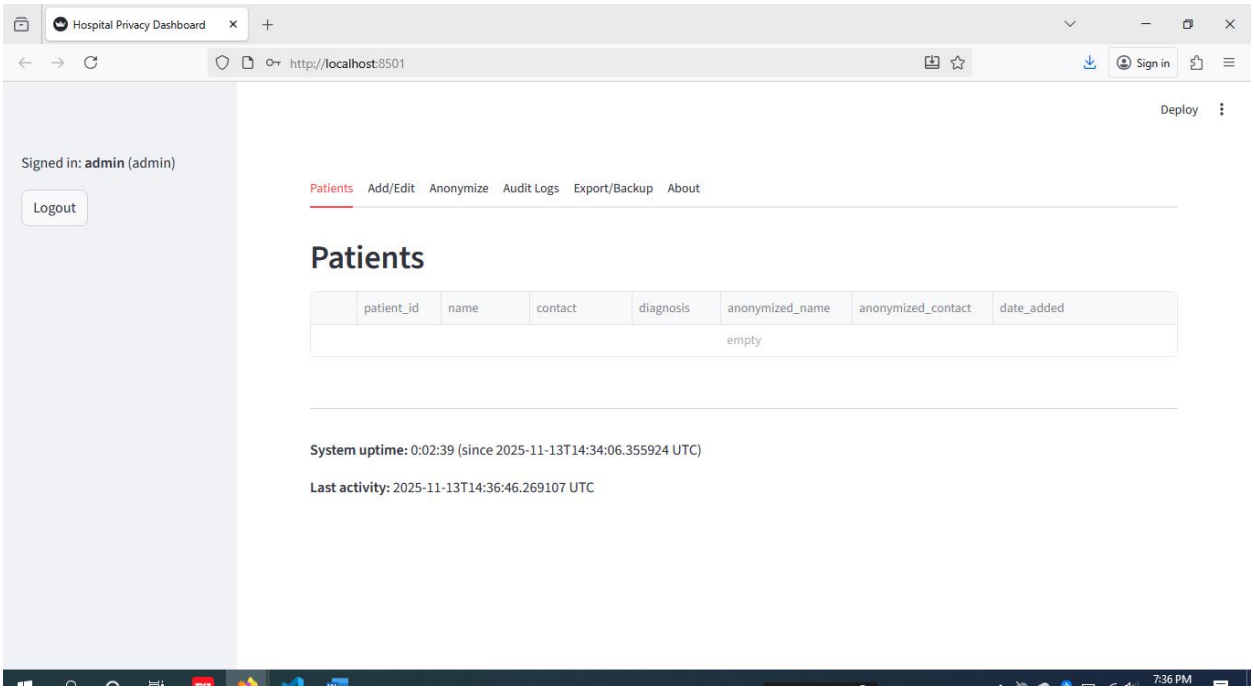


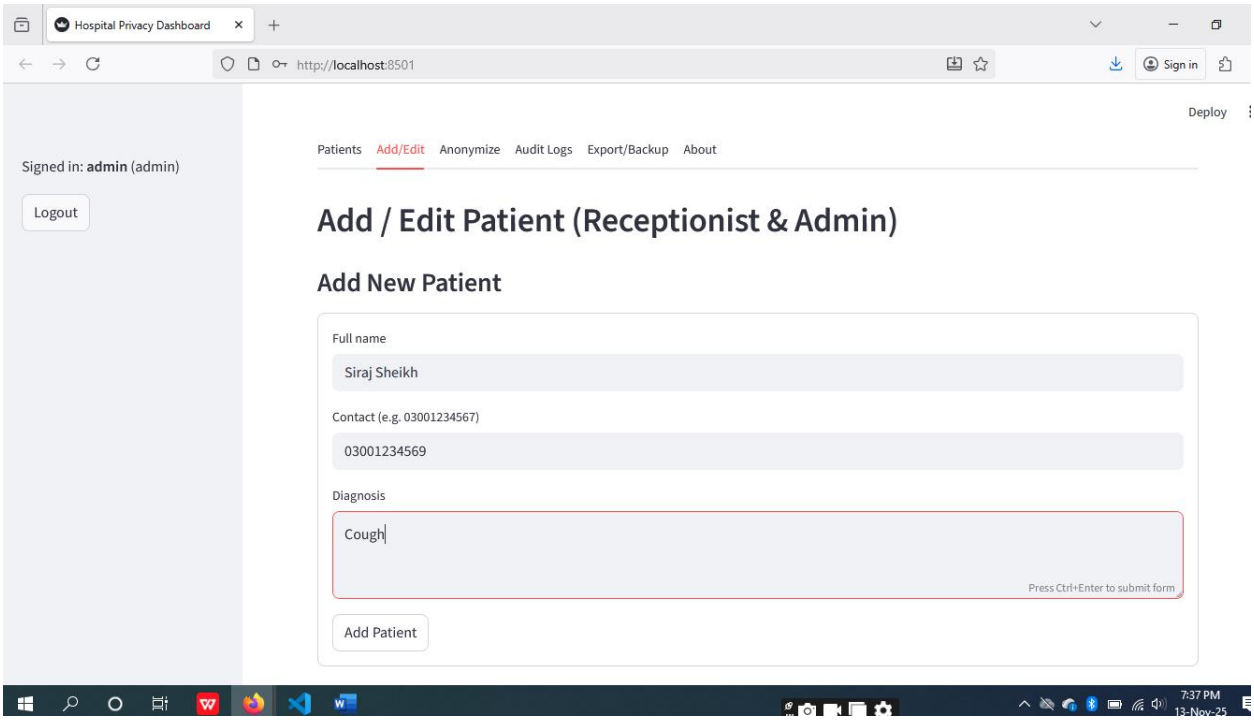

Hospital Privacy Dashboard
http://localhost:8501
Sign in
Deploy
Hospital Management Dashboard — Login
Username
admin
Password
Login
Login successful! Loading dashboard...

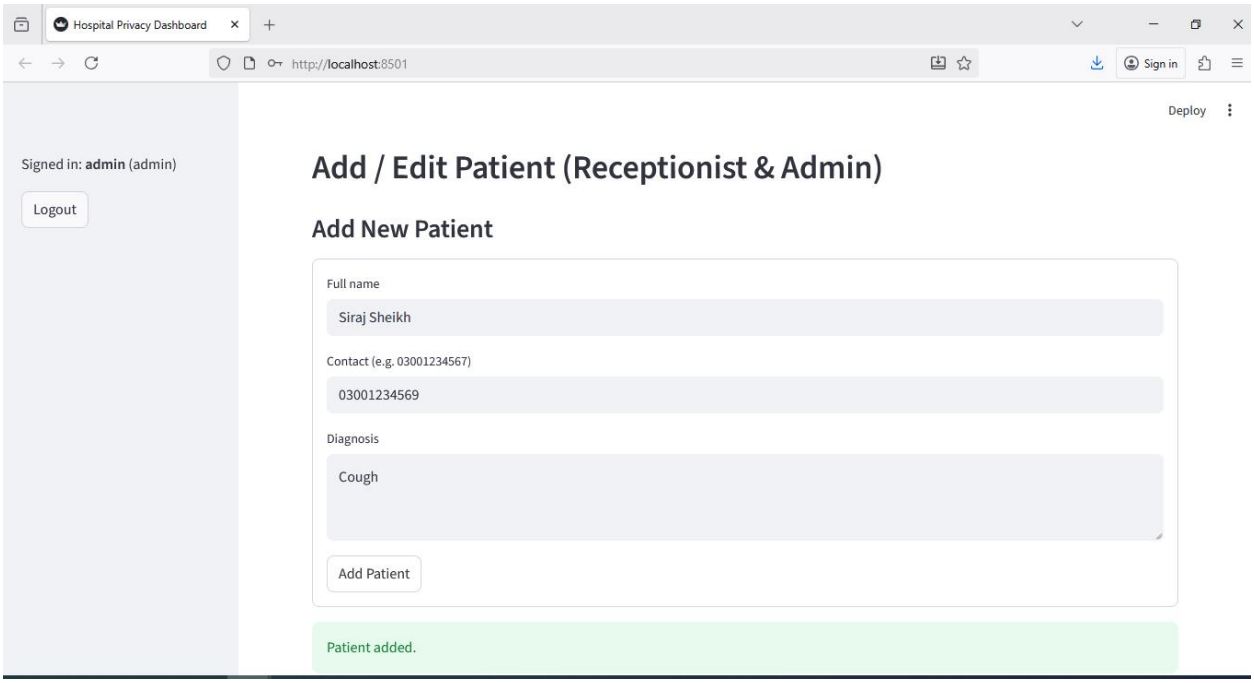
User = Admin:

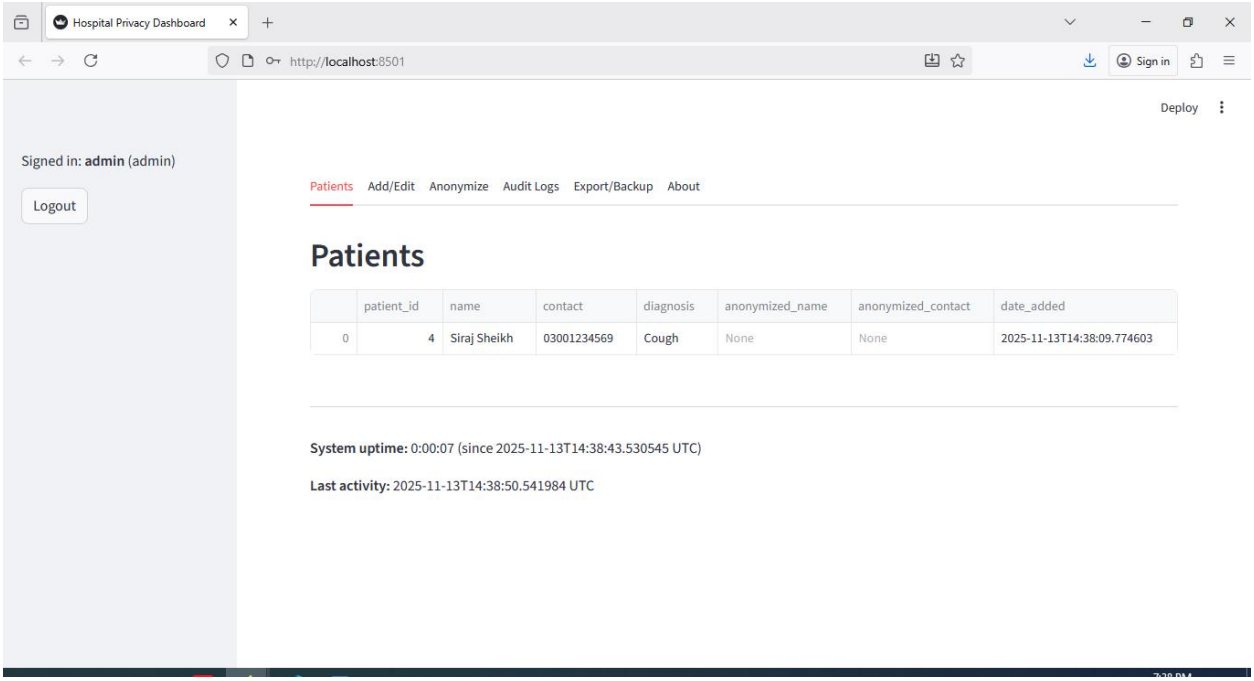

Hospital Privacy Dashboard
http://localhost:8501
Sign in
Deploy
Hospital Management Dashboard — Login
Username
Password
Login
Footer: Uptime: 0:00:00
7:35 PM
13-Nov-25

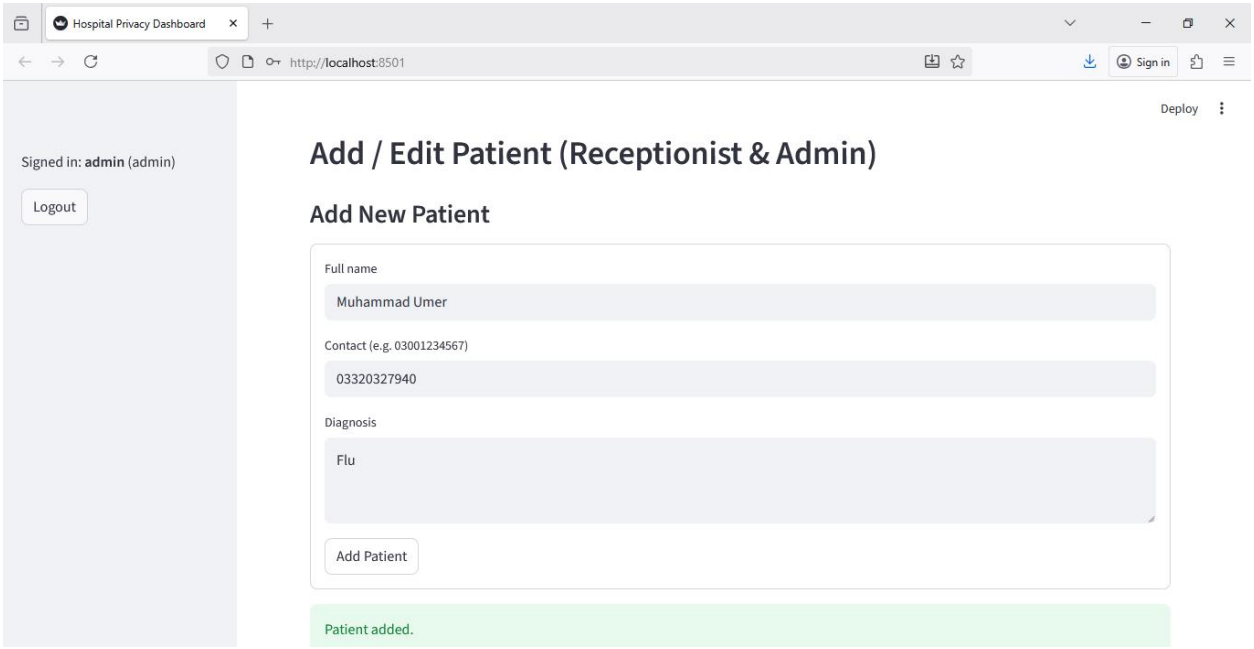

Hospital Privacy Dashboard
http://localhost:8501
Sign in
Deploy
Hospital Management Dashboard — Login
Username
admin
Password
Press Enter to submit form
Login
Footer: Uptime: 0:00:00
7:36 PM
13-Nov-25


Hospital Privacy Dashboard
http://localhost:8501
Sign in
Deploy
Signed in: admin (admin)
Logout
Patients
Add/Edit
Anonymize
Audit Logs
Export/Backup
About
Patients
patient_id
name
contact
diagnosis
anonymized_name
anonymized_contact
date_added
empty
System uptime: 0:02:39 (since 2025-11-13T14:34:06.355924 UTC)
Last activity: 2025-11-13T14:36:46.269107 UTC
7:36 PM


Hospital Privacy Dashboard
http://localhost:8501
Sign in
Deploy
Signed in: admin (admin)
Logout
Patients
Add/Edit
Anonymize
Audit Logs
Export/Backup
About
Add / Edit Patient (Receptionist & Admin)
Add New Patient
Full name
Siraj Sheikh
Contact (e.g. 03001234567)
03001234569
Diagnosis
Cough
Press Ctrl+Enter to submit form
Add Patient
7:37 PM
13-Nov-25


Hospital Privacy Dashboard
http://localhost:8501
Sign in
Deploy
Signed in: admin (admin)
Logout
Add / Edit Patient (Receptionist & Admin)
Add New Patient
Full name
Siraj Sheikh
Contact (e.g. 03001234567)
03001234569
Diagnosis
Cough
Add Patient
Patient added.


Hospital Privacy Dashboard
http://localhost:8501
Sign in
Deploy
Signed in: admin (admin)
Logout
Patients
Add/Edit
Anonymize
Audit Logs
Export/Backup
About
Patients
patient_id
name
contact
diagnosis
anonymized_name
anonymized_contact
date_added
0
4
Siraj Sheikh
03001234569
Cough
None
None
2025-11-13T14:38:09.774603
System uptime: 0:00:07 (since 2025-11-13T14:38:43.530545 UTC)
Last activity: 2025-11-13T14:38:50.541984 UTC


Hospital Privacy Dashboard
http://localhost:8501
Sign in
Deploy
Signed in: admin (admin)
Logout
Add / Edit Patient (Receptionist & Admin)
Add New Patient
Full name
Muhammad Umer
Contact (e.g. 03001234567)
03320327940
Diagnosis
Flu
Add Patient
Patient added.

Add / Edit Patient (Receptionist & Admin)

Add New Patient

Full name

Subhan

Contact (e.g. 03001234567)

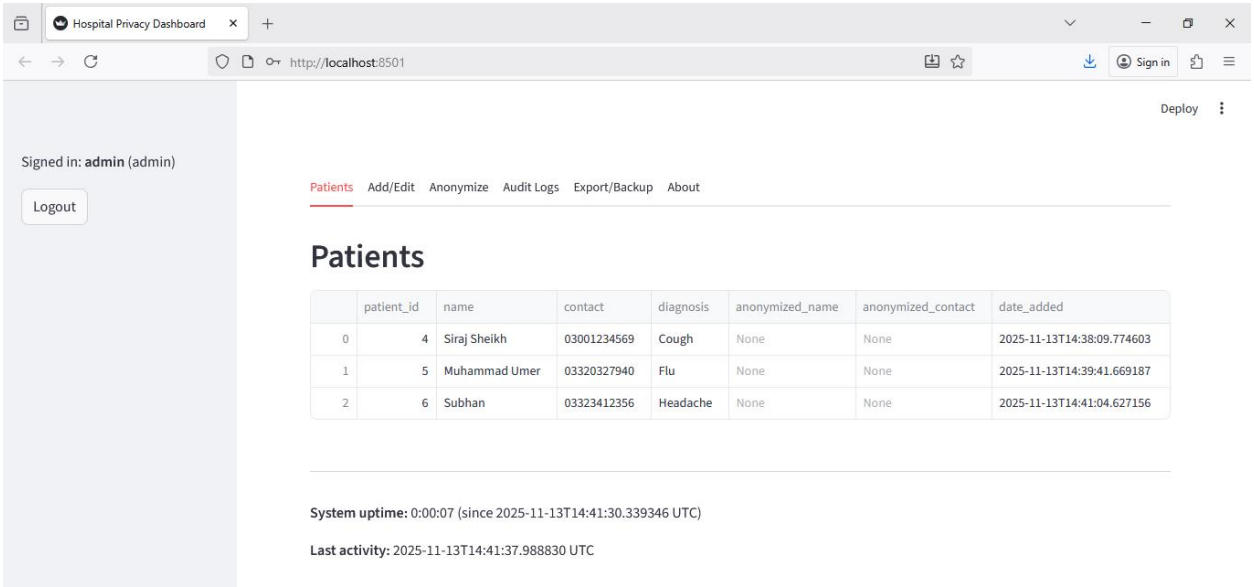
03323412356

Diagnosis

Headache

Add Patient

Patient added.



Patients Add/Edit Anonymize Audit Logs Export/Backup About

Patients

	patient_id	name	contact	diagnosis	anonymized_name	anonymized_contact	date_added
0	4	Siraj Sheikh	03001234569	Cough	None	None	2025-11-13T14:38:09.774603
1	5	Muhammad Umer	03320327940	Flu	None	None	2025-11-13T14:39:41.669187
2	6	Subhan	03323412356	Headache	None	None	2025-11-13T14:41:04.627156

System uptime: 0:00:07 (since 2025-11-13T14:41:30.339346 UTC)

Last activity: 2025-11-13T14:41:37.988830 UTC

Edit Existing Patient

Select Patient to Edit

4 - Siraj Sheikh

Name

Siraj Sheikh

Contact

03001234569

Diagnosis

light Cough

Update Record

Edit Existing Patient

Select Patient to Edit

4 - Siraj Sheikh

Name

Siraj Sheikh

Contact

03001234569

Diagnosis

light Cough

Update Record

Patient record updated successfully.

http://localhost:8501

Sign in

Deploy

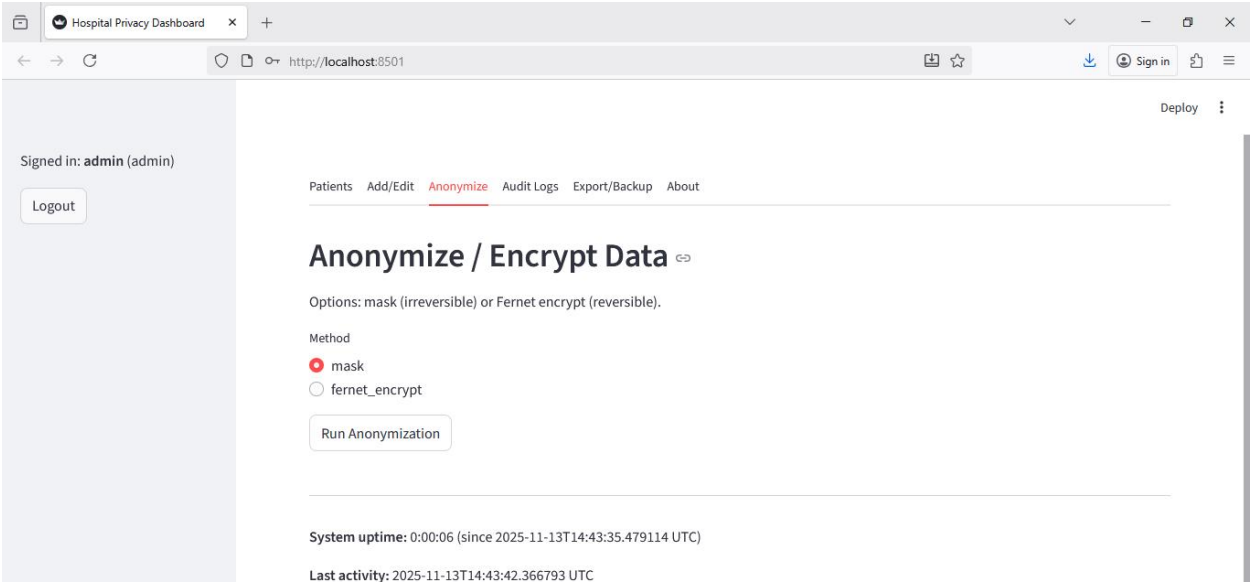
Patients Add/Edit Anonymize Audit Logs Export/Backup About

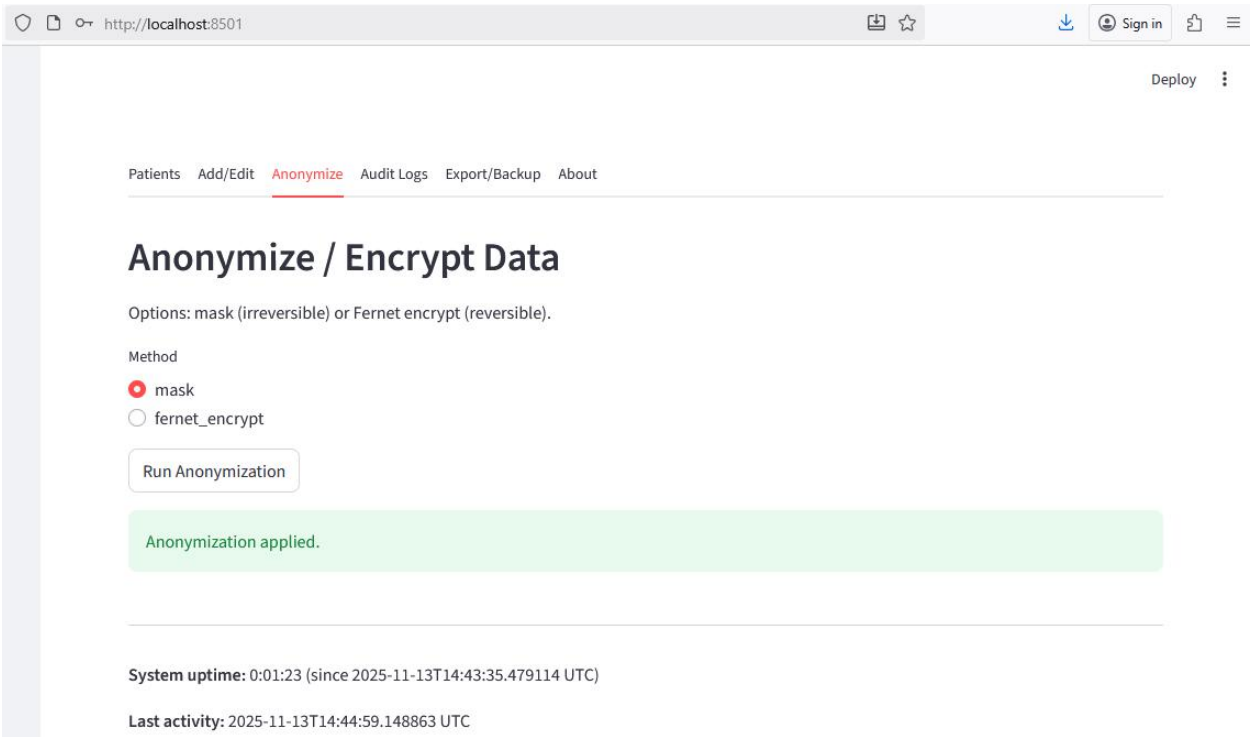
Patients

	patient_id	name	contact	diagnosis	anonymized_name	anonymized_contact	date_added
0	4	Siraj Sheikh	03001234569	light Cough	None	None	2025-11-13T14:38:09.774603
1	5	Muhammad Umer	03320327940	Flu	None	None	2025-11-13T14:39:41.669187
2	6	Subhan	03323412356	Headache	None	None	2025-11-13T14:41:04.627156

System uptime: 0:00:06 (since 2025-11-13T14:43:35.479114 UTC)

Last activity: 2025-11-13T14:43:42.366793 UTC


Hospital Privacy Dashboard
http://localhost:8501
Sign in
Deploy
Signed in: admin (admin)
Logout
Patients
Add/Edit
Anonymize
Audit Logs
Export/Backup
About
Anonymize / Encrypt Data
Options: mask (irreversible) or Fernet encrypt (reversible).
Method
mask
fernet_encrypt
Run Anonymization
System uptime: 0:00:06 (since 2025-11-13T14:43:35.479114 UTC)
Last activity: 2025-11-13T14:43:42.366793 UTC


http://localhost:8501
Sign in
Deploy
Patients
Add/Edit
Anonymize
Audit Logs
Export/Backup
About
Anonymize / Encrypt Data
Options: mask (irreversible) or Fernet encrypt (reversible).
Method
mask
fernet_encrypt
Run Anonymization
Anonymization applied.
System uptime: 0:01:23 (since 2025-11-13T14:43:35.479114 UTC)
Last activity: 2025-11-13T14:44:59.148863 UTC

http://localhost:8501

Sign in

Deploy

Patients Add/Edit Anonymize Audit Logs Export/Backup About

Patients

	patient_id	name	contact	diagnosis	anonymized_name	anonymized_contact	date_added
0	4	Siraj Sheikh	03001234569	light Cough	ANON_d96559	XXX-XXX-4569	2025-11-13T14:38:09.774603
1	5	Muhammad Umer	03320327940	Flu	ANON_130021	XXX-XXX-7940	2025-11-13T14:39:41.669187
2	6	Subhan	03323412356	Headache	ANON_46c41b	XXX-XXX-2356	2025-11-13T14:41:04.627156

System uptime: 0:00:09 (since 2025-11-13T14:45:37.251614 UTC)

Last activity: 2025-11-13T14:45:46.686892 UTC

http://localhost:8501

Sign in

Deploy

Patients Add/Edit Anonymize Audit Logs Export/Backup About

Audit / Integrity Logs

	log_id	user_id	role	action	timestamp	details
0	99	1	admin	view_patients	2025-11-13T14:45:46.666618	rows=3
1	98	1	admin	login	2025-11-13T14:45:46.607019	admin logged in
2	97	1	admin	view_logs	2025-11-13T14:44:59.148863	count=96
3	96	1	admin	anonymize	2025-11-13T14:44:59.135126	method=mask, rows=3
4	95	1	admin	view_patients	2025-11-13T14:44:59.096581	rows=3
5	94	1	admin	view_logs	2025-11-13T14:43:42.366793	count=93
6	93	1	admin	view_patients	2025-11-13T14:43:42.339864	rows=3
7	92	1	admin	login	2025-11-13T14:43:42.279039	admin logged in
8	91	1	admin	view_logs	2025-11-13T14:43:03.536785	count=90
9	90	1	admin	update_patient	2025-11-13T14:43:03.521038	Updated patient ID 4

3	96	1	admin	anonymize	2025-11-13T14:44:59.135126	method=mask, rows=3
4	95	1	admin	view_patients	2025-11-13T14:44:59.096581	rows=3
5	94	1	admin	view_logs	2025-11-13T14:43:42.366793	count=93
6	93	1	admin	view_patients	2025-11-13T14:43:42.339864	rows=3
7	92	1	admin	login	2025-11-13T14:43:42.279039	admin logged in
8	91	1	admin	view_logs	2025-11-13T14:43:03.536785	count=90
9	90	1	admin	update_patient	2025-11-13T14:43:03.521038	Updated patient ID 4

Export logs CSV

System uptime: 0:00:09 (since 2025-11-13T14:45:37.251614 UTC)

Last activity: 2025-11-13T14:45:46.686892 UTC

http://localhost:8501

Patients Add/Edit Anonymize Audit Logs Export/Backup About

Export / Backup

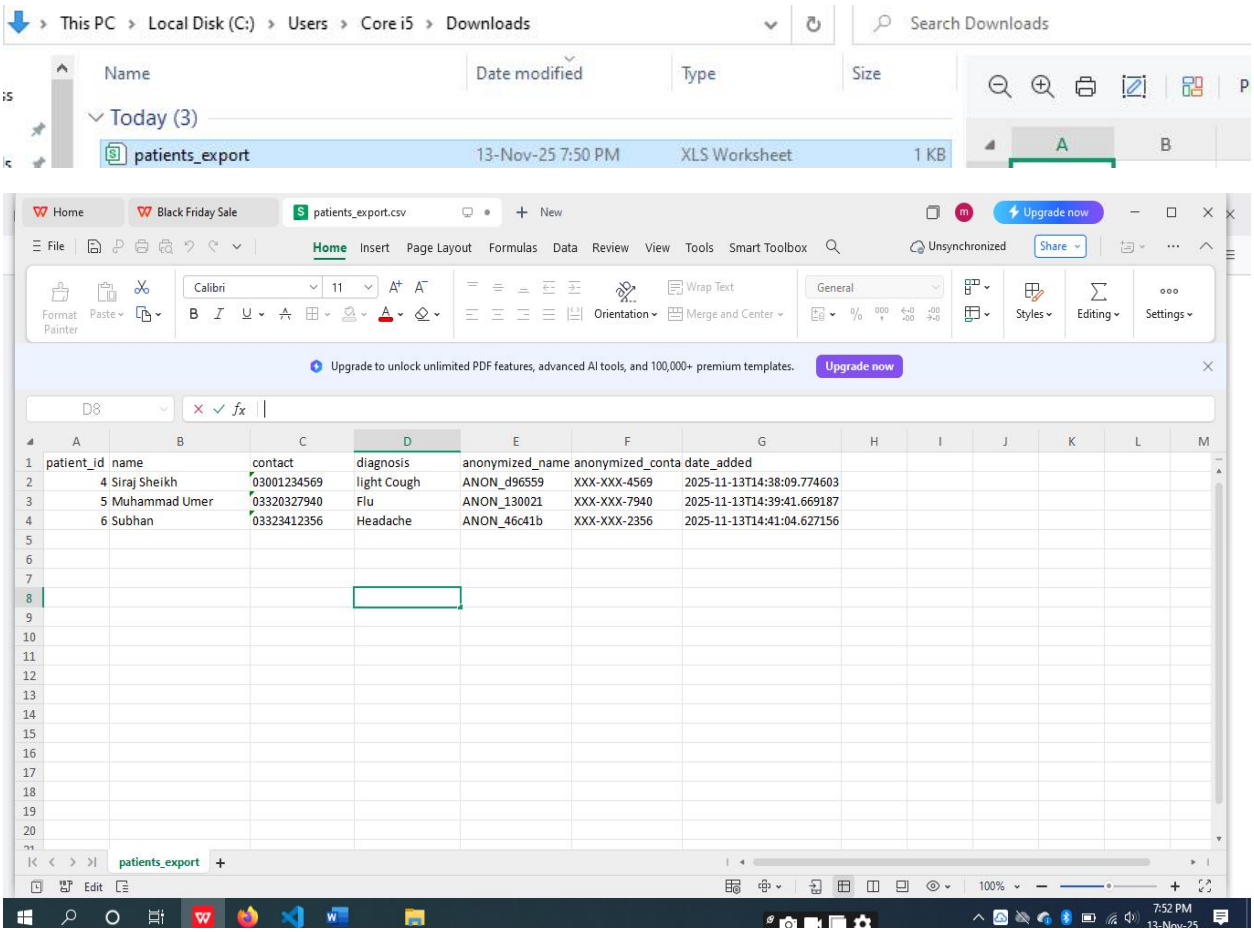
Download patient data (respecting role restrictions).

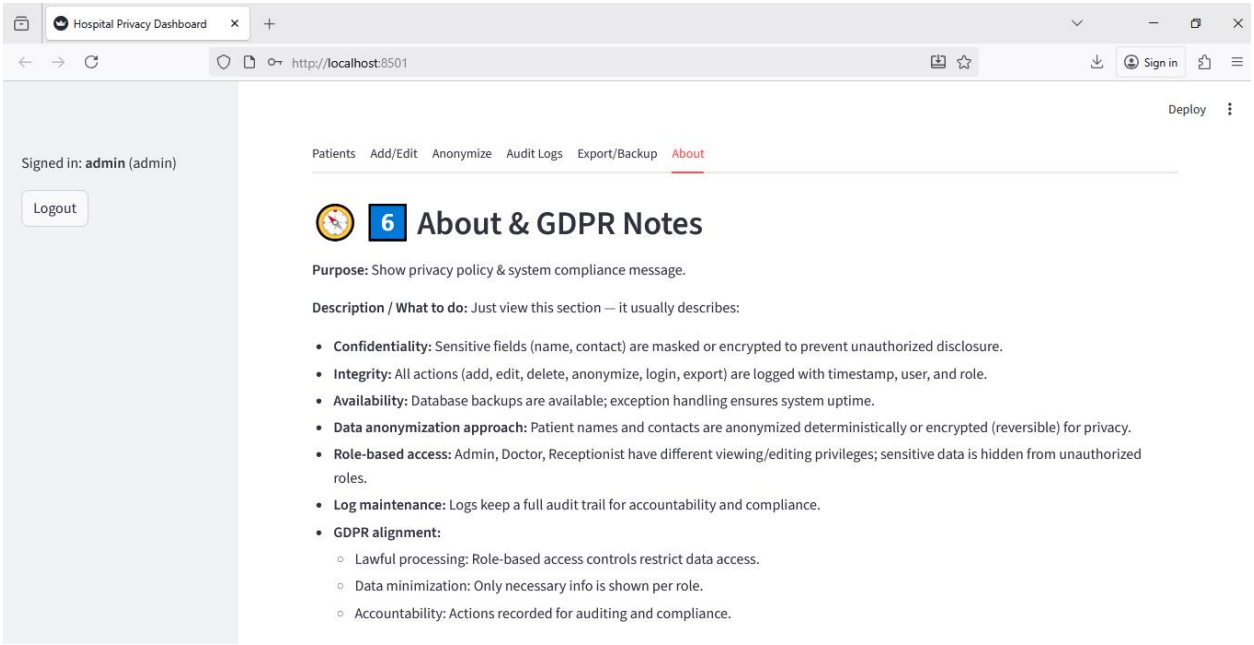
Download Patients CSV

Backup DB (create copy)

System uptime: 0:00:09 (since 2025-11-13T14:45:37.251614 UTC)

Last activity: 2025-11-13T14:45:46.686892 UTC


This PC › Local Disk (C:) › Users › Core i5 › Downloads
Search Downloads
Name
Date modified
Type
Size
Today (3)
patients_export
13-Nov-25 7:50 PM
XLS Worksheet
1 KB
patients_export.csv
Upgrade to unlock unlimited PDF features, advanced AI tools, and 100,000+ premium templates.
Upgrade now
patient_id name contact diagnosis anonymized_name anonymized_conta date_added
4 Siraj Sheikh 03001234569 light Cough ANON_d96559 XXX-XXX-4569 2025-11-13T14:38:09.774603
5 Muhammad Umer 03320327940 Flu ANON_130021 XXX-XXX-7940 2025-11-13T14:39:41.669187
6 Subhan 03323412356 Headache ANON_46c41b XXX-XXX-2356 2025-11-13T14:41:04.627156
patients_export


Hospital Privacy Dashboard
http://localhost:8501
Signed in: admin (admin)
Logout
Deploy
Patients Add/Edit Anonymize Audit Logs Export/Backup About
6 About & GDPR Notes
Purpose: Show privacy policy & system compliance message.
Description / What to do: Just view this section — it usually describes:
Confidentiality: Sensitive fields (name, contact) are masked or encrypted to prevent unauthorized disclosure.
Integrity: All actions (add, edit, delete, anonymize, login, export) are logged with timestamp, user, and role.
Availability: Database backups are available; exception handling ensures system uptime.
Data anonymization approach: Patient names and contacts are anonymized deterministically or encrypted (reversible) for privacy.
Role-based access: Admin, Doctor, Receptionist have different viewing/editing privileges; sensitive data is hidden from unauthorized roles.
Log maintenance: Logs keep a full audit trail for accountability and compliance.
GDPR alignment:
Lawful processing: Role-based access controls restrict data access.
Data minimization: Only necessary info is shown per role.
Accountability: Actions recorded for auditing and compliance.

Purpose: Show privacy policy & system compliance message.

Description / What to do: Just view this section — it usually describes:

- **Confidentiality:** Sensitive fields (name, contact) are masked or encrypted to prevent unauthorized disclosure.
- **Integrity:** All actions (add, edit, delete, anonymize, login, export) are logged with timestamp, user, and role.
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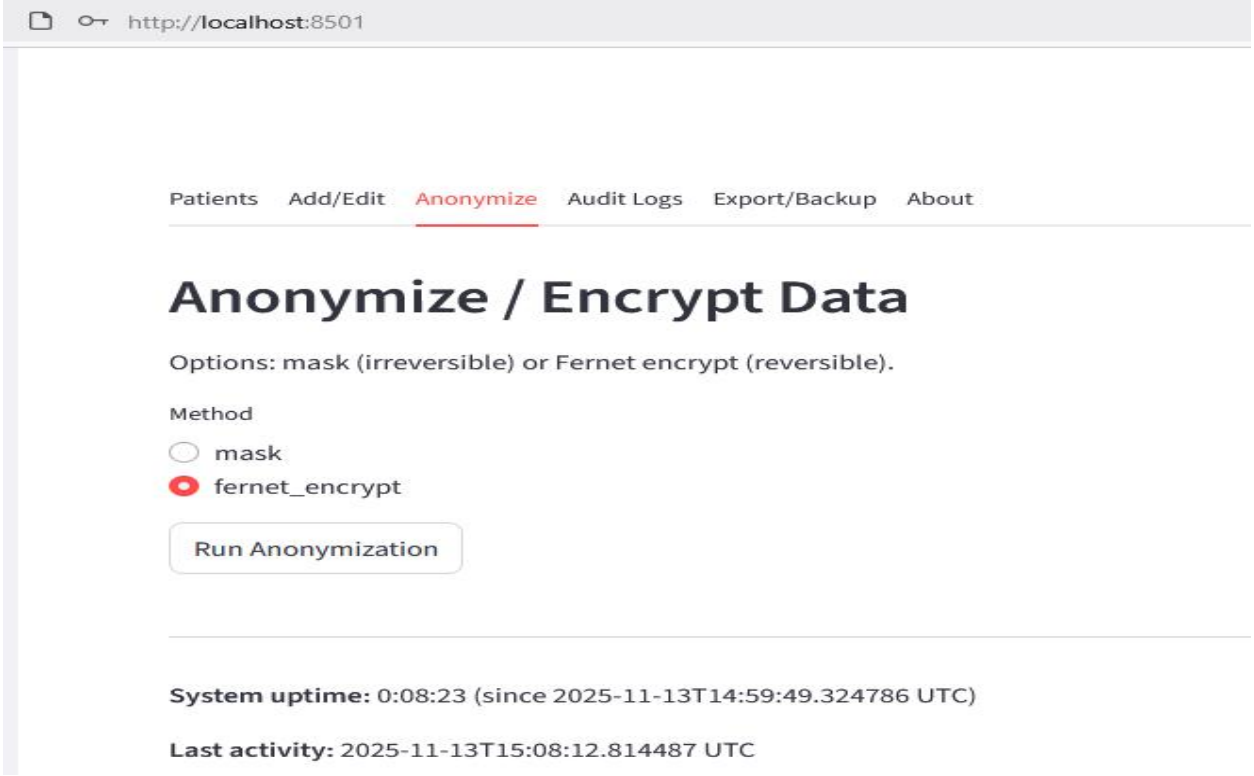
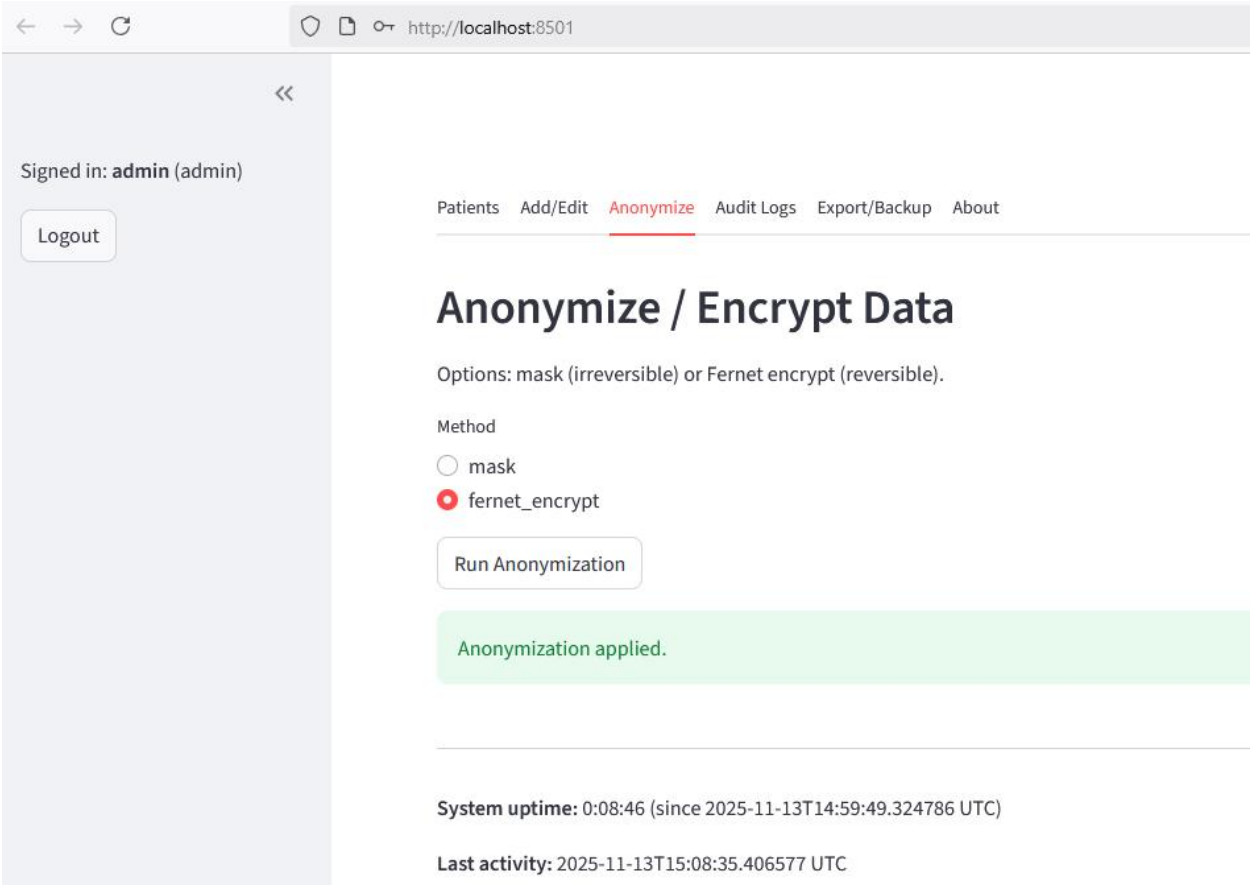
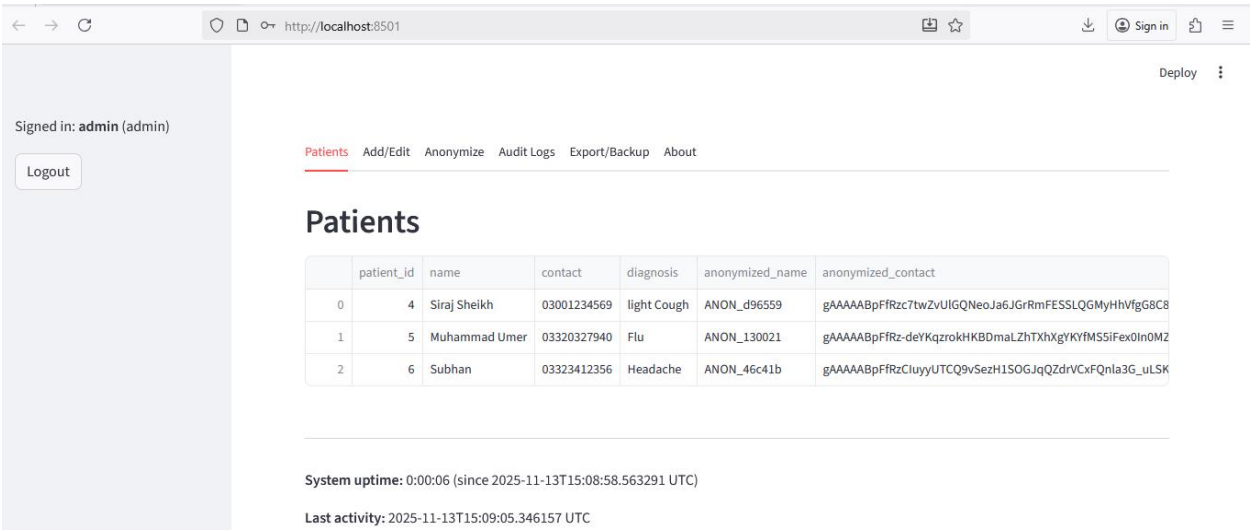
- **Log maintenance:** Logs keep a full audit trail for accountability and compliance.
- **GDPR alignment:**
  - Lawful processing: Role-based access controls restrict data access.
  - Data minimization: Only necessary info is shown per role.
  - Accountability: Actions recorded for auditing and compliance.

System uptime: 0:00:06 (since 2025-11-13T14:59:49.324786 UTC)

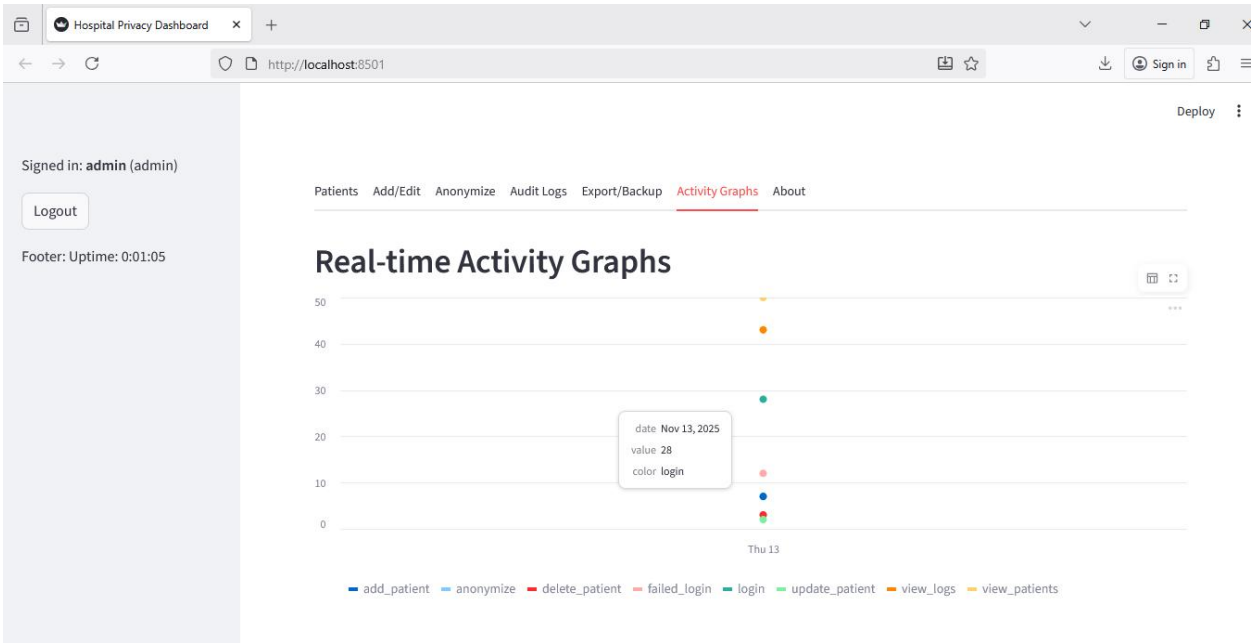
Last activity: 2025-11-13T14:59:56.302815 UTC

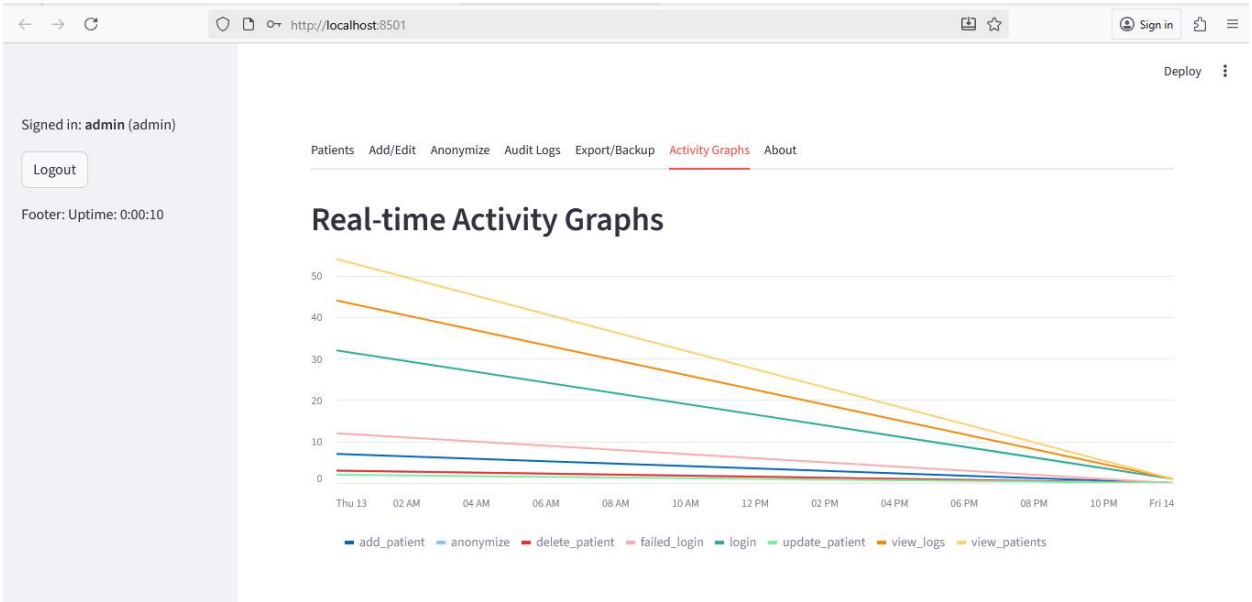




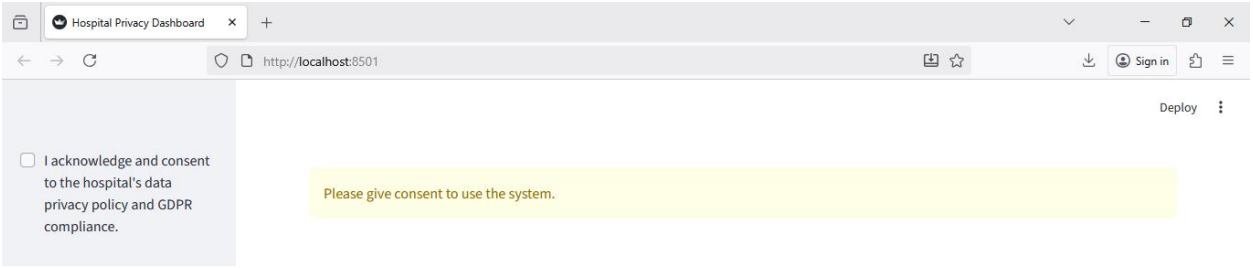


	patient_id	name	contact	diagnosis	anonymized_name	anonymized_contact
0	4	Siraj Sheikh	03001234569	light Cough	ANON_d96559	gAAAAABpFfRzc7twZvUlGQNeoJa6JGrRmFESSLQGMyHhVfgG8C8
1	5	Muhammad Umer	03320327940	Flu	ANON_130021	gAAAAABpFfRz-deYKqzrokHKBDmaLZhTXhXgYKYfM5SiFex0In0M2
2	6	Subhan	03323412356	Headache	ANON_46c41b	gAAAAABpFfRzCIuyyUTCQ9vSezH1SOGJqQZdrVCxFQnla3G_uLSK


Hospital Privacy Dashboard
http://localhost:8501
Sign in
Deploy
Signed in: admin (admin)
Logout
Footer: Uptime: 0:01:05
Patients
Add/Edit
Anonymize
Audit Logs
Export/Backup
Activity Graphs
About
Real-time Activity Graphs
50
40
30
20
10
0
date Nov 13, 2025
value 28
color login
Thu 13
add_patient
anonymize
delete_patient
failed_login
login
update_patient
view_logs
view_patients


http://localhost:8501
Sign in
Deploy
Signed in: admin (admin)
Logout
Footer: Uptime: 0:00:10
Patients
Add/Edit
Anonymize
Audit Logs
Export/Backup
Activity Graphs
About
Real-time Activity Graphs
50
40
30
20
10
0
Thu 13
02 AM
04 AM
06 AM
08 AM
10 AM
12 PM
02 PM
04 PM
06 PM
08 PM
10 PM
Fri 14
add_patient
anonymize
delete_patient
failed_login
login
update_patient
view_logs
view_patients

User = Doctor:



Hospital Management Dashboard — Login

Username

drbob

Password

••••••

Login

Login successful! Loading dashboard...

Hospital Management Dashboard — Login

Username

drbob

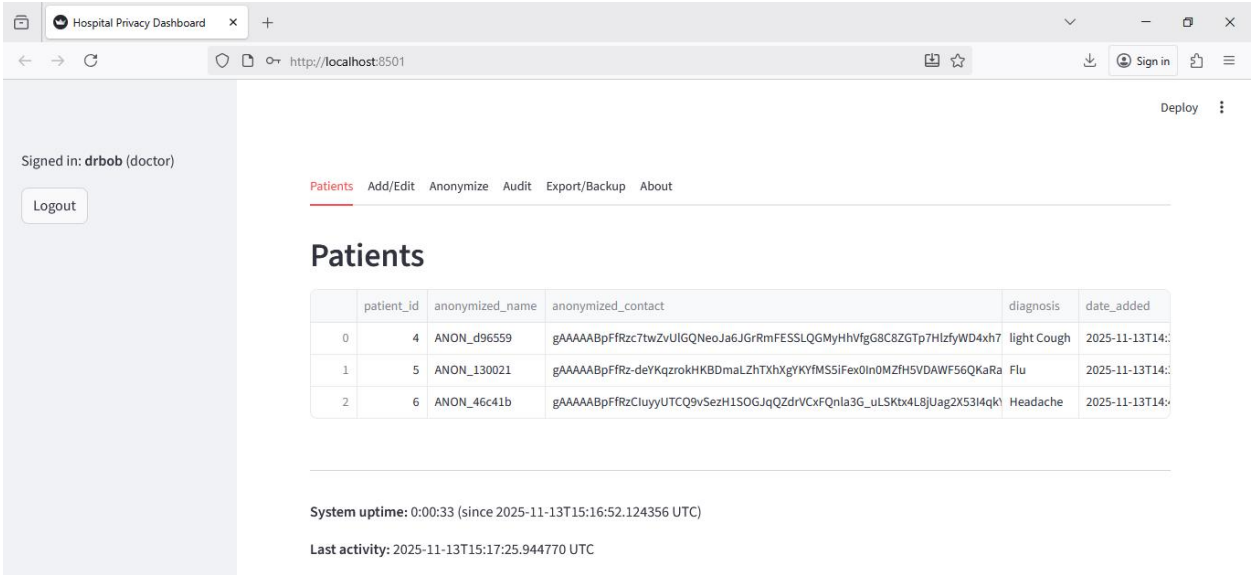
Password

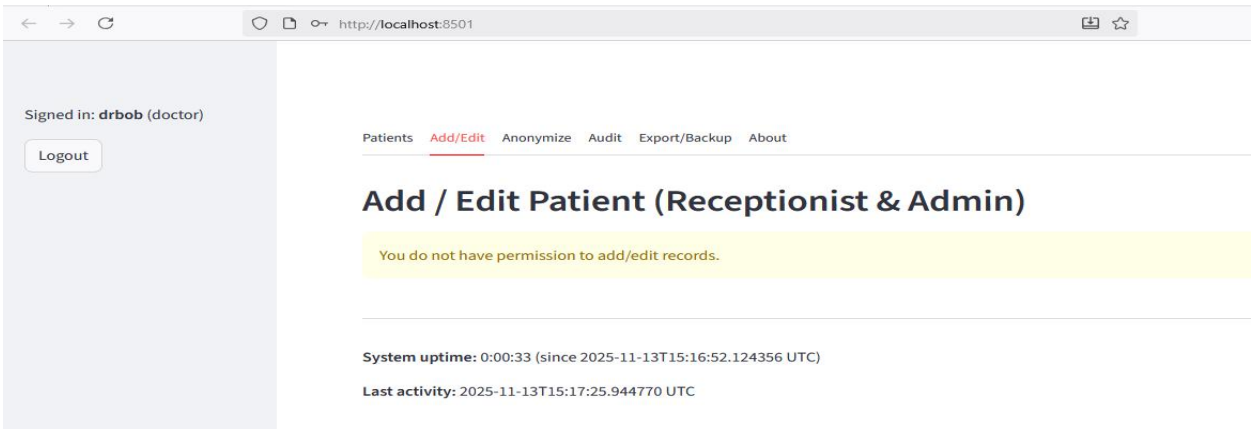
••••••

Press Enter to submit form

Login

Footer: Uptime: 0:00:00


Hospital Privacy Dashboard
http://localhost:8501
Sign in
Deploy
Signed in: drbob (doctor)
Logout
Patients Add/Edit Anonymize Audit Export/Backup About
Patients
patient_id anonymized_name anonymized_contact diagnosis date_added
0 4 ANON_d96559 gAAAAABpFfRzc7twZvUlGQNeoJa6JGrRmFESSLQGMyHhVfgG8C8ZGTp7HlzfyWD4xh7 light Cough 2025-11-13T14:
1 5 ANON_130021 gAAAAABpFfRz-deYKqzrokHKBDmaLZhTXhXgYKYfMS5iFex0In0MZfH5VDAWF56QKaRa Flu 2025-11-13T14:
2 6 ANON_46c41b gAAAAABpFfRzCIuyyUTCQ9vSezH1SOGJqQZdrVCxFQnla3G_uLSKtx4L8jUag2X53I4qk Headache 2025-11-13T14:
System uptime: 0:00:33 (since 2025-11-13T15:16:52.124356 UTC)
Last activity: 2025-11-13T15:17:25.944770 UTC


http://localhost:8501
Signed in: drbob (doctor)
Logout
Patients Add/Edit Anonymize Audit Export/Backup About
Add / Edit Patient (Receptionist & Admin)
You do not have permission to add/edit records.
System uptime: 0:00:33 (since 2025-11-13T15:16:52.124356 UTC)
Last activity: 2025-11-13T15:17:25.944770 UTC

Signed in: **drbob** (doctor)

Logout

Patients Add/Edit Anonymize Audit Export/Backup About

Anonymize / Encrypt Data

Only Admin can run anonymization.

System uptime: 0:00:33 (since 2025-11-13T15:16:52.124356 UTC)

Last activity: 2025-11-13T15:17:25.944770 UTC

http://localhost:8501

Signed in: **drbob** (doctor)

Logout

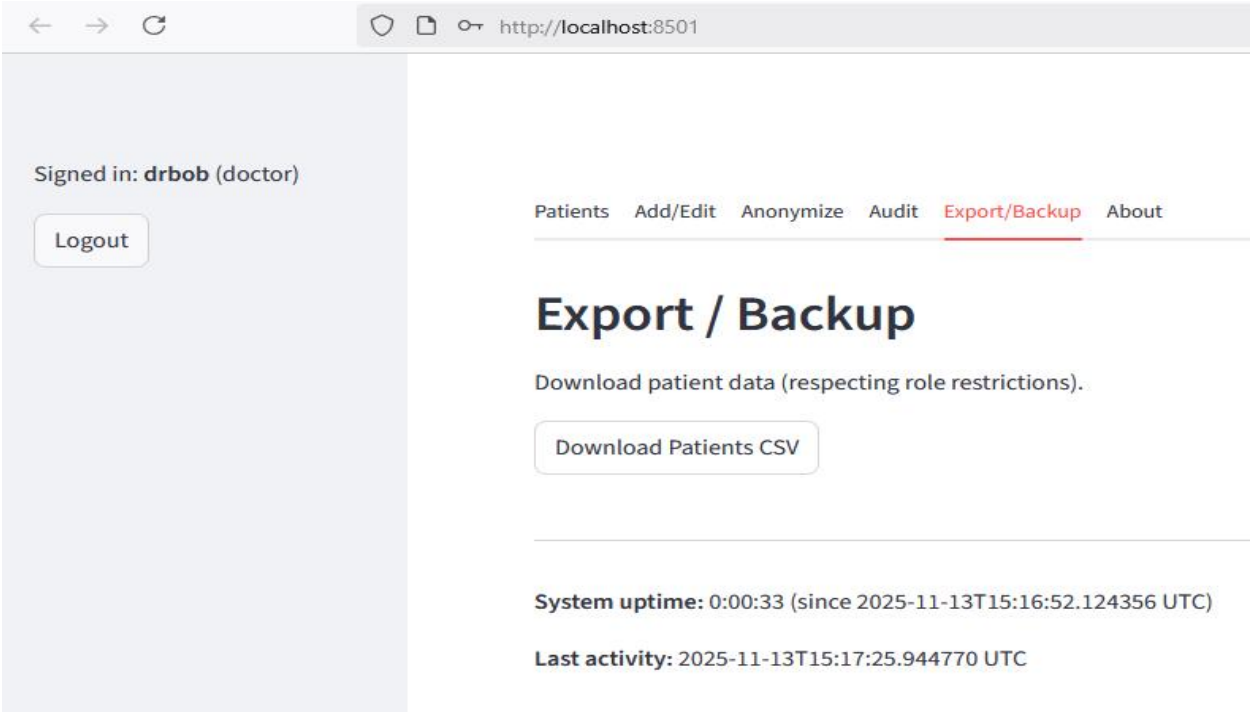
Patients Add/Edit Anonymize Audit Export/Backup About

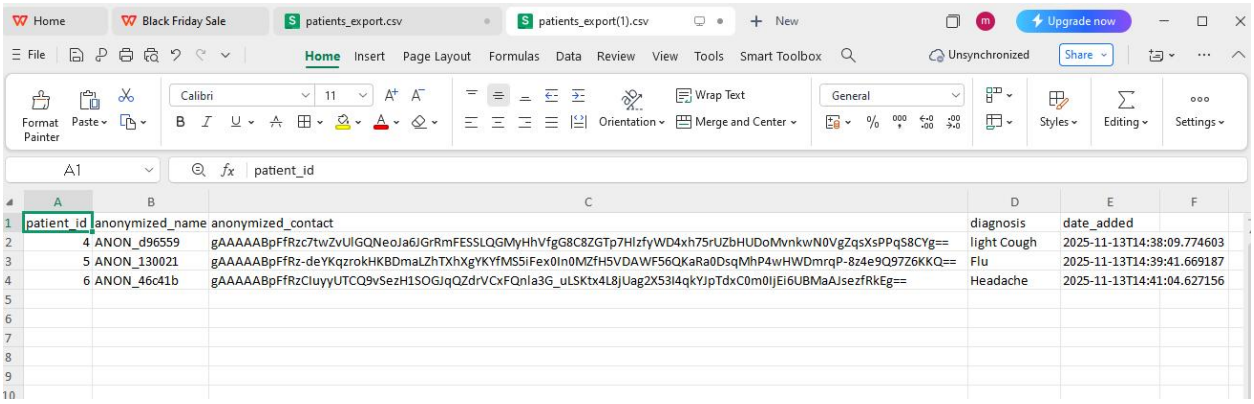
Audit / Integrity Logs

Audit logs view restricted to Admin.

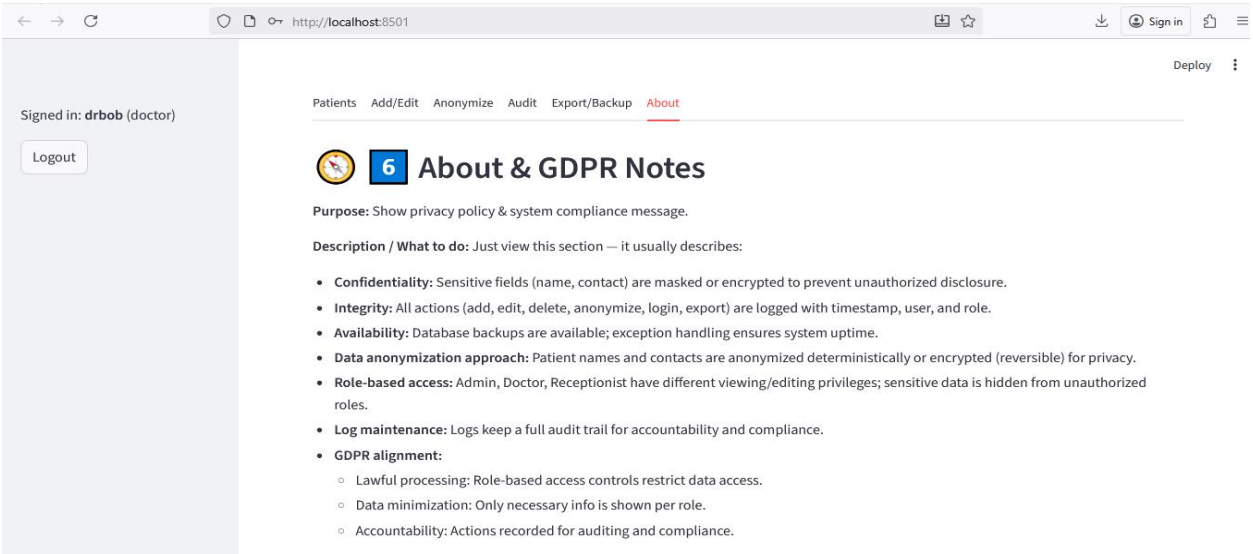
System uptime: 0:00:33 (since 2025-11-13T15:16:52.124356 UTC)

Last activity: 2025-11-13T15:17:25.944770 UTC





patient_id	anonymized_name	anonymized_contact	diagnosis	date_added
4	ANON_d96559	gAAAAABpFfRzc7twZvUlGQNeoJa6JGrRmFESSLQGMyHhVfgG8C8ZGTp7HlzfyWD4xh75rUZbHUDoMvnkwN0VgZqsXsPPqS8CYg==	light Cough	2025-11-13T14:38:09.774603
5	ANON_130021	gAAAAABpFfRz-deYKqzrokHKBDmaLZhTXhXgYKYfMS5iFex0In0MZfH5VDAWF56QKaRa0DsqMhP4wHWDmrqP-8z4e9Q97Z6KKQ==	Flu	2025-11-13T14:39:41.669187
6	ANON_46c41b	gAAAAABpFfRzCIuyyUTCQ9vSezH1SOGJqQZdrVCxFQnla3G_uLSKtx4L8jUag2X53I4qkYJpTdxC0m0IjEi6UBMaAJsezfRkEg==	Headache	2025-11-13T14:41:04.627156



Signed in: **drbob** (doctor)

Logout

Description / What to do: Just view this section — it usually describes:

- **Confidentiality:** Sensitive fields (name, contact) are masked or encrypted to prevent unauthorized disclosure.
- **Integrity:** All actions (add, edit, delete, anonymize, login, export) are logged with timestamp, user, and role.
- **Availability:** Database backups are available; exception handling ensures system uptime.
- **Data anonymization approach:** Patient names and contacts are anonymized deterministically or encrypted (reversible) for privacy.
- **Role-based access:** Admin, Doctor, Receptionist have different viewing/editing privileges; sensitive data is hidden from unauthorized roles.
- **Log maintenance:** Logs keep a full audit trail for accountability and compliance.
- **GDPR alignment:**
  - Lawful processing: Role-based access controls restrict data access.
  - Data minimization: Only necessary info is shown per role.
  - Accountability: Actions recorded for auditing and compliance.

System uptime: 0:02:15 (since 2025-11-13T15:16:52.124356 UTC)

Last activity: 2025-11-13T15:19:07.306384 UTC

8:20 PM

Hospital Privacy Dashboard

http://localhost:8501

Sign in

Deploy

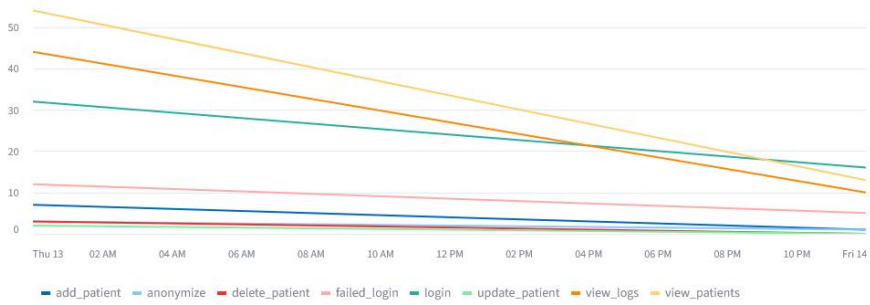
Signed in: **drbob** (doctor)

Logout

Footer: Uptime: 0:01:47

Patients | Add/Edit | Anonymize | Audit | Export/Backup | Activity Graphs | About & GDPR Notes

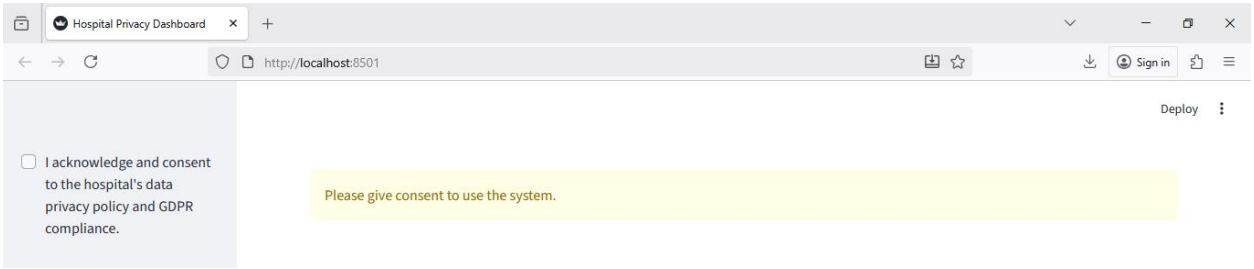
Real-time Activity Graphs

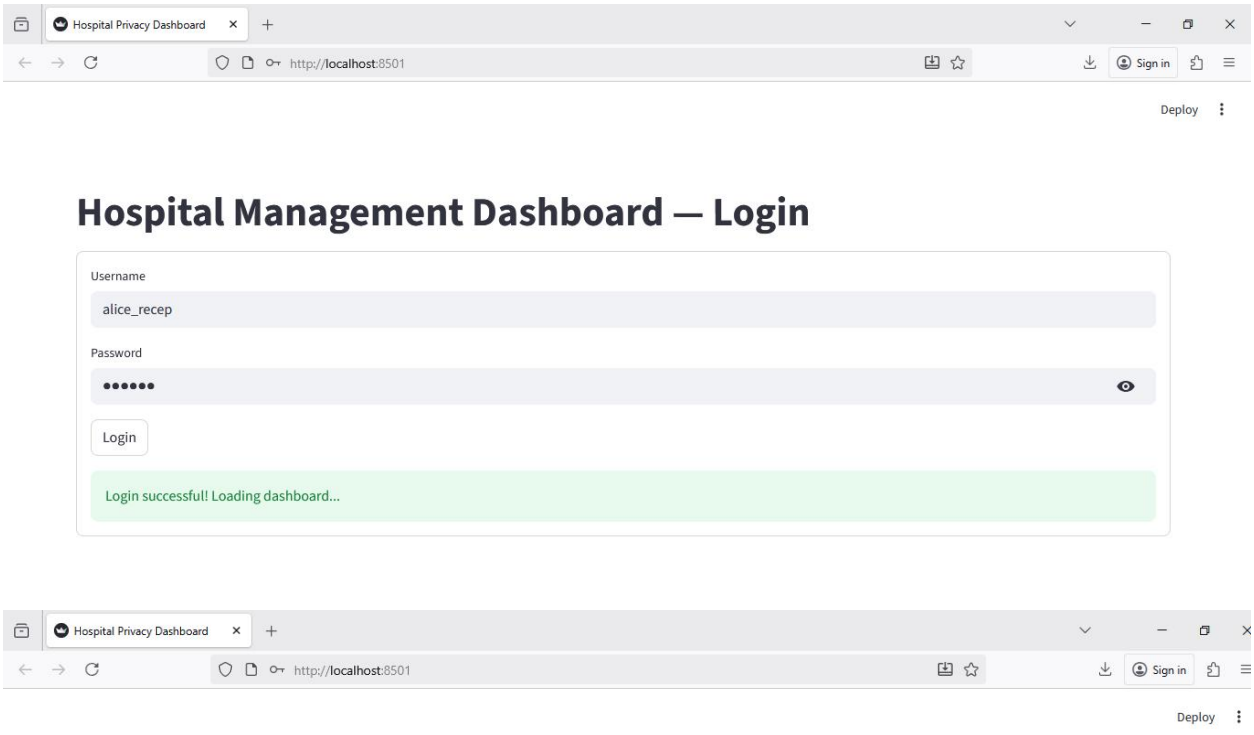


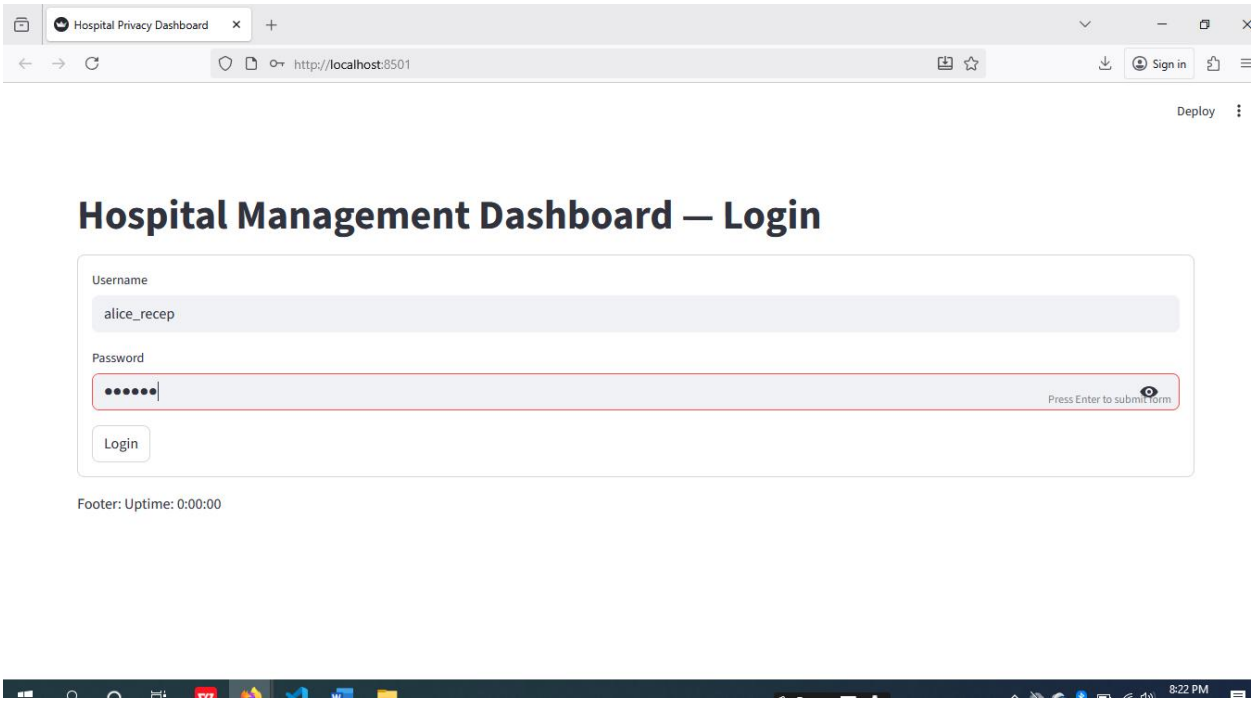
System uptime: 0:01:47 (since 2025-11-14T06:35:06.499219 UTC)

Last activity: 2025-11-14T06:36:54.261553 UTC

User = Receptionalist:







Hospital Privacy Dashboard

http://localhost:8501

Sign in

Deploy

Signed in: **alice_recep** (receptionist)

Logout

Patients Add/Edit Anonymize Audit Export/Backup About

Patients

	patient_id	date_added	diagnosis
0	4	2025-11-13T14:38:09.774603	light Cough
1	5	2025-11-13T14:39:41.669187	Flu
2	6	2025-11-13T14:41:04.627156	Headache

System uptime: 0:01:14 (since 2025-11-13T15:21:11.397611 UTC)

Last activity: 2025-11-13T15:22:26.214500 UTC

http://localhost:8501

Signed in: **alice_recep** (receptionist)

Logout

Add / Edit Patient (Receptionist & Admin)

Add New Patient

Full name

Daniyal

Contact (e.g. 03001234567)

03214561235

Diagnosis

fever

Add Patient

Patient added.

Deploy

Signed in: **alice_recep** (receptionist)

Logout

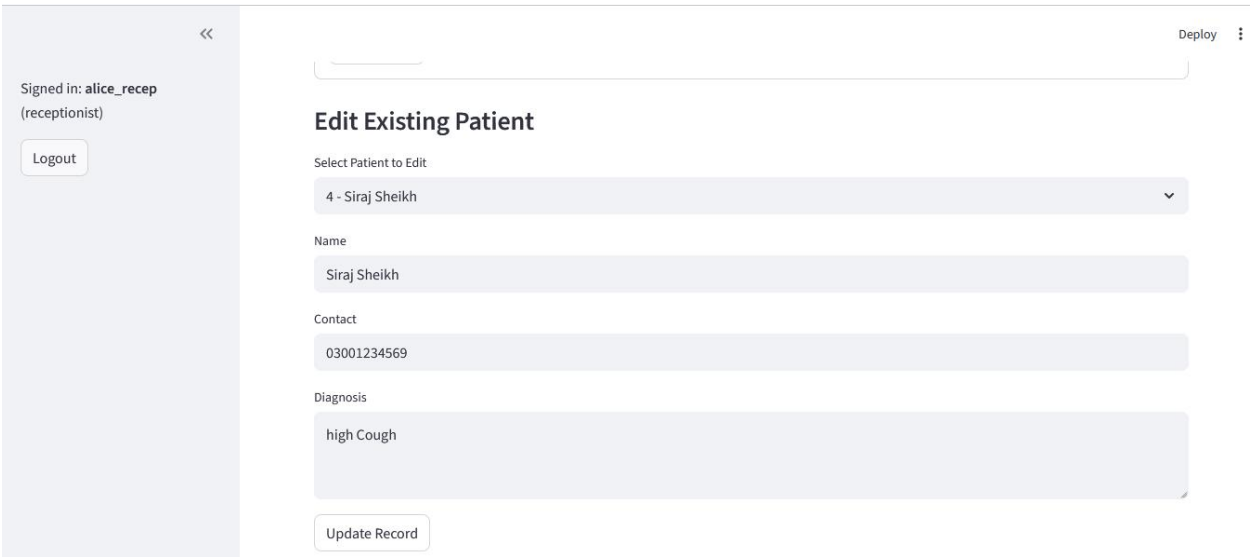
Patients Add/Edit Anonymize Audit Export/Backup About

Patients

	patient_id	date_added	diagnosis
0	4	2025-11-13T14:38:09.774603	light Cough
1	5	2025-11-13T14:39:41.669187	Flu
2	6	2025-11-13T14:41:04.627156	Headache
3	7	2025-11-13T15:27:51.390616	fever

System uptime: 0:07:22 (since 2025-11-13T15:21:11.397611 UTC)

Last activity: 2025-11-13T15:28:33.651897 UTC


Signed in: alice_recep
(receptionist)
Logout
Deploy
Edit Existing Patient
Select Patient to Edit
4 - Siraj Sheikh
Name
Siraj Sheikh
Contact
03001234569
Diagnosis
high Cough
Update Record

Signed in: alice_recep
(receptionist)
Logout
Patients
Add/Edit
Anonymize
Audit
Export/Backup
About
Anonymize / Encrypt Data
Only Admin can run anonymization.
System uptime: 0:07:22 (since 2025-11-13T15:21:11.397611 UTC)
Last activity: 2025-11-13T15:28:33.651897 UTC

Signed in: **alice_recep**
(receptionist)

Logout

Patients Add/Edit Anonymize Audit Export/Backup About

Audit / Integrity Logs

Audit logs view restricted to Admin.

System uptime: 0:07:22 (since 2025-11-13T15:21:11.397611 UTC)

Last activity: 2025-11-13T15:28:33.651897 UTC

Signed in: **alice_recep**
(receptionist)

Logout

Patients Add/Edit Anonymize Audit Export/Backup About

Export / Backup

Download patient data (respecting role restrictions).

Download Patients CSV

System uptime: 0:07:22 (since 2025-11-13T15:21:11.397611 UTC)

Last activity: 2025-11-13T15:28:33.651897 UTC

Signed in: **alice_recep** (receptionist)

Logout

Patients Add/Edit Anonymize Audit Export/Backup About

🧭 6 About & GDPR Notes

Purpose: Show privacy policy & system compliance message.

Description / What to do: Just view this section — it usually describes:

- **Confidentiality:** Sensitive fields (name, contact) are masked or encrypted to prevent unauthorized disclosure.
- **Integrity:** All actions (add, edit, delete, anonymize, login, export) are logged with timestamp, user, and role.
- **Availability:** Database backups are available; exception handling ensures system uptime.
- **Data anonymization approach:** Patient names and contacts are anonymized deterministically or encrypted (reversible) for privacy.
- **Role-based access:** Admin, Doctor, Receptionist have different viewing/editing privileges; sensitive data is hidden from unauthorized roles.
- **Log maintenance:** Logs keep a full audit trail for accountability and compliance.
- **GDPR alignment:**
  - Lawful processing: Role-based access controls restrict data access.
  - Data minimization: Only necessary info is shown per role.
  - Accountability: Actions recorded for auditing and compliance.

Deploy ⋮

Signed in: **admin** (admin)

Logout

Patients Add/Edit Anonymize Audit Logs Export/Backup About

Patients

	patient_id	name	contact	diagnosis	anonymized_name	anonymized_contact
0	4	Siraj Sheikh	03001234569	light Cough	ANON_d96559	gAAAAABpFfRzc7twZvUlGQNeoJa6JGrRmFESSLQGMyHhVfgG8C8
1	5	Muhammad Umer	03320327940	Flu	ANON_130021	gAAAAABpFfRz-deYKqzrokHKBDmaLZhTXhXgYKYfMS5iFex0In0MZ
2	6	Subhan	03323412356	Headache	ANON_46c41b	gAAAAABpFfRzCIuyyUTCQ9vSezH1SOGJqQZdrVCxFQnla3G_uLSK
3	7	Daniyal	03214561235	fever	None	None

System uptime: 0:00:08 (since 2025-11-13T15:31:11.070032 UTC)

Last activity: 2025-11-13T15:31:19.332911 UTC

http://localhost:8501

Sign in

Deploy

Signed in: **admin** (admin)

Logout

Patients Add/Edit Anonymize Audit Logs Export/Backup About

Audit / Integrity Logs

	log_id	user_id	role	action	timestamp	details
0	133	1	admin	view_patients	2025-11-13T15:34:41.779473	rows=4
1	132	1	admin	login	2025-11-13T15:34:41.711120	admin logged in
2	131	3	receptionist	view_patients	2025-11-13T15:33:16.827411	rows=4
3	130	3	receptionist	login	2025-11-13T15:33:16.760051	alice_recep logged in
4	129	None	unknown	failed_login	2025-11-13T15:32:59.947993	username=alice_recep
5	128	1	admin	view_logs	2025-11-13T15:31:19.332911	count=127
6	127	1	admin	view_patients	2025-11-13T15:31:19.308917	rows=4
7	126	1	admin	login	2025-11-13T15:31:19.243170	admin logged in
8	125	3	receptionist	view_patients	2025-11-13T15:28:33.651897	rows=4
9	124	3	receptionist	add_patient	2025-11-13T15:27:51.398652	name=Daniyal

Signed in: **admin** (admin)

Logout

3	130	3	receptionist	login	2025-11-13T15:33:16.760051	alice_recep logged in
4	129	None	unknown	failed_login	2025-11-13T15:32:59.947993	username=alice_recep
5	128	1	admin	view_logs	2025-11-13T15:31:19.332911	count=127
6	127	1	admin	view_patients	2025-11-13T15:31:19.308917	rows=4
7	126	1	admin	login	2025-11-13T15:31:19.243170	admin logged in
8	125	3	receptionist	view_patients	2025-11-13T15:28:33.651897	rows=4
9	124	3	receptionist	add_patient	2025-11-13T15:27:51.398652	name=Daniyal

Export logs CSV

System uptime: 0:00:05 (since 2025-11-13T15:34:36.195592 UTC)

Last activity: 2025-11-13T15:34:41.803257 UTC

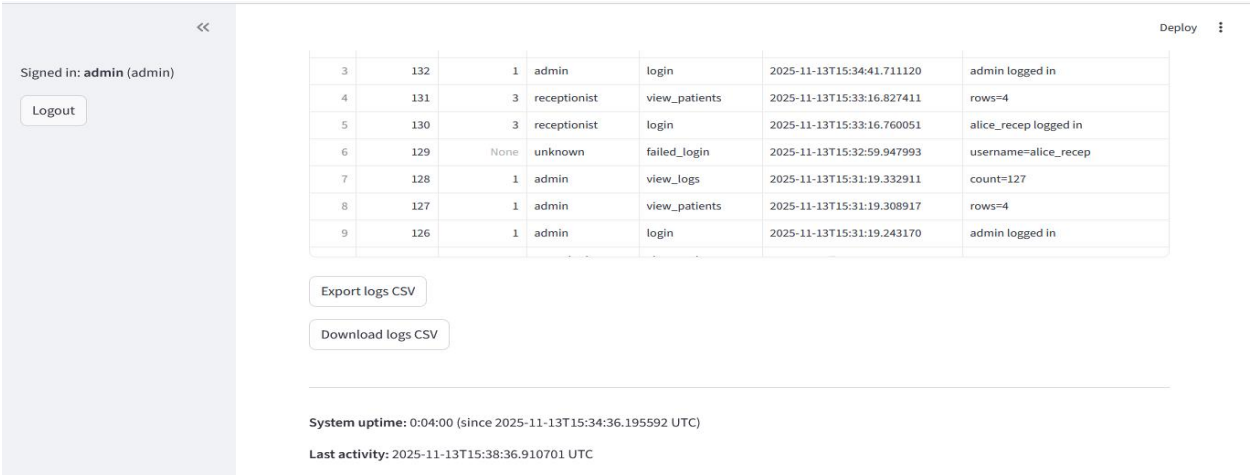
Signed in: **admin** (admin)

Logout

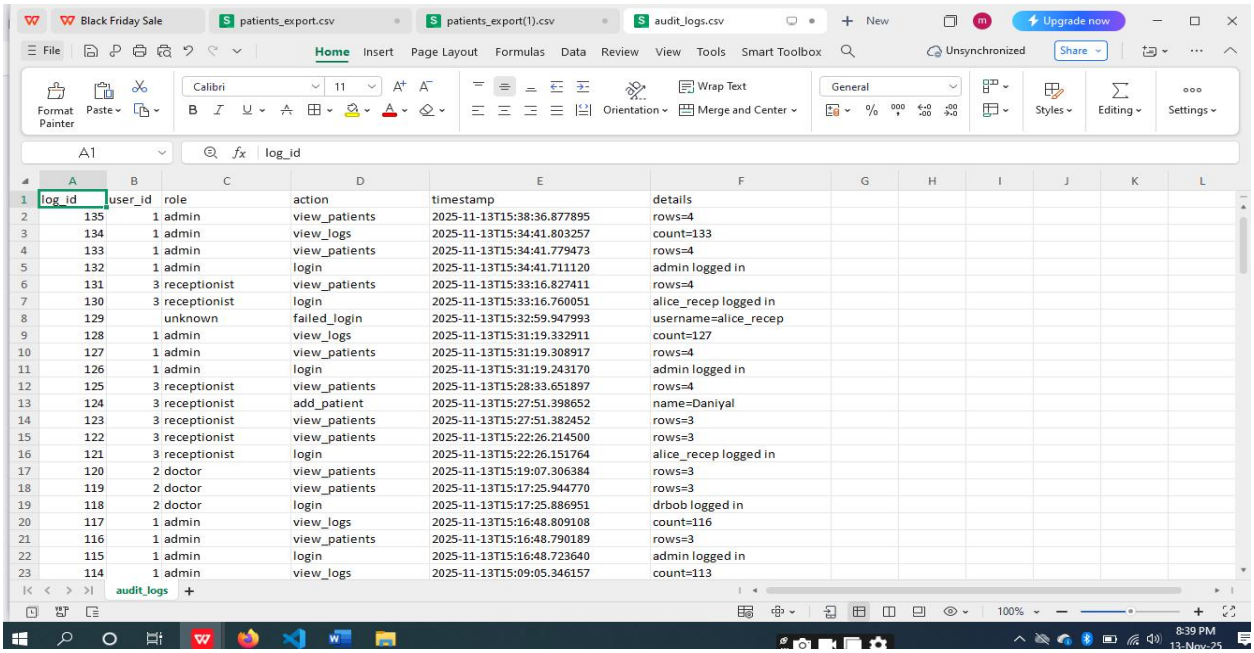
Depl

Audit / Integrity Logs

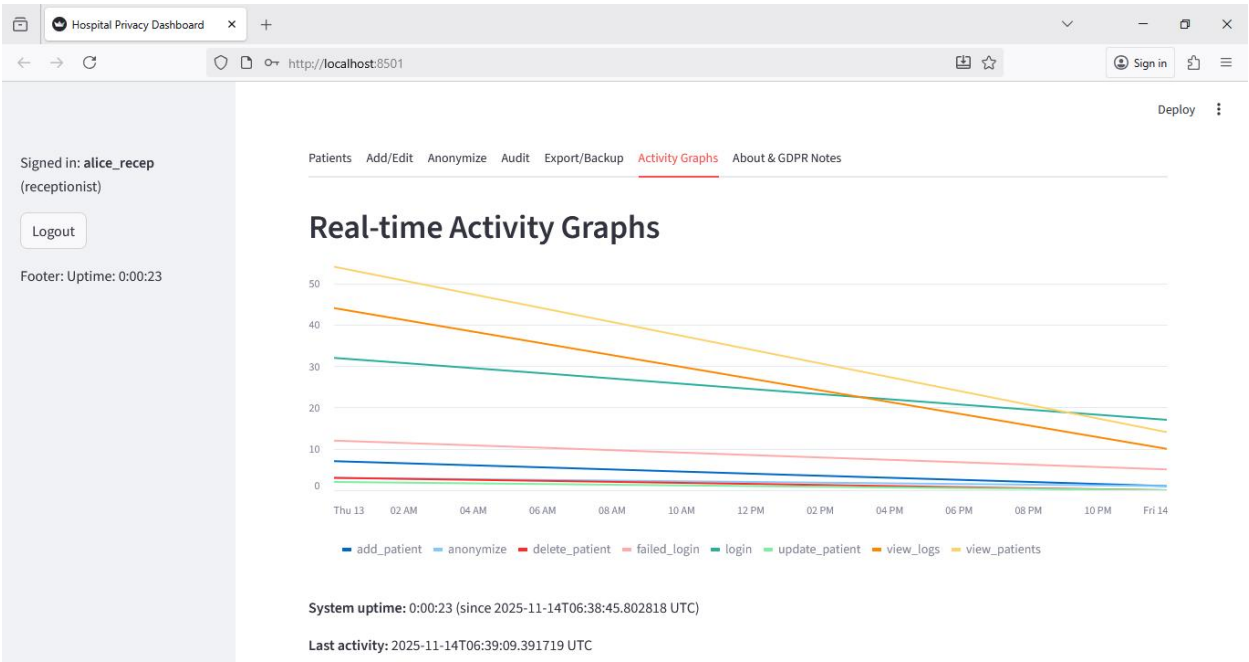
	log_id	user_id	role	action	timestamp	details
29	104	1	admin	login	2025-11-13T14:59:56.214161	admin logged in
30	103	None	unknown	failed_login	2025-11-13T14:59:52.836428	username=admin
31	102	1	admin	view_logs	2025-11-13T14:50:23.508375	count=101
32	101	1	admin	view_patients	2025-11-13T14:50:23.422086	rows=3
33	100	1	admin	view_logs	2025-11-13T14:45:46.686892	count=99
34	99	1	admin	view_patients	2025-11-13T14:45:46.666618	rows=3
35	98	1	admin	login	2025-11-13T14:45:46.607019	admin logged in
36	97	1	admin	view_logs	2025-11-13T14:44:59.148863	count=96
37	96	1	admin	anonymize	2025-11-13T14:44:59.135126	method=mask, rows=3
38	95	1	admin	view_patients	2025-11-13T14:44:59.096581	rows=3



3	132	1	admin	login	2025-11-13T15:34:41.711120	admin logged in
4	131	3	receptionist	view_patients	2025-11-13T15:33:16.827411	rows=4
5	130	3	receptionist	login	2025-11-13T15:33:16.760051	alice_recep logged in
6	129	None	unknown	failed_login	2025-11-13T15:32:59.947993	username=alice_recep
7	128	1	admin	view_logs	2025-11-13T15:31:19.332911	count=127
8	127	1	admin	view_patients	2025-11-13T15:31:19.308917	rows=4
9	126	1	admin	login	2025-11-13T15:31:19.243170	admin logged in



log_id	user_id	role	action	timestamp	details
135	1	admin	view_patients	2025-11-13T15:38:36.877895	rows=4
134	1	admin	view_logs	2025-11-13T15:34:41.803257	count=133
133	1	admin	view_patients	2025-11-13T15:34:41.779473	rows=4
132	1	admin	login	2025-11-13T15:34:41.711120	admin logged in
131	3	receptionist	view_patients	2025-11-13T15:33:16.827411	rows=4
130	3	receptionist	login	2025-11-13T15:33:16.760051	alice_recep logged in
129		unknown	failed_login	2025-11-13T15:32:59.947993	username=alice_recep
128	1	admin	view_logs	2025-11-13T15:31:19.332911	count=127
127	1	admin	view_patients	2025-11-13T15:31:19.308917	rows=4
126	1	admin	login	2025-11-13T15:31:19.243170	admin logged in
125	3	receptionist	view_patients	2025-11-13T15:28:33.651897	rows=4
124	3	receptionist	add_patient	2025-11-13T15:27:51.398652	name=Daniyal
123	3	receptionist	view_patients	2025-11-13T15:27:51.382452	rows=3
122	3	receptionist	view_patients	2025-11-13T15:22:26.214500	rows=3
121	3	receptionist	login	2025-11-13T15:22:26.151764	alice_recep logged in
120	2	doctor	view_patients	2025-11-13T15:19:07.306384	rows=3
119	2	doctor	view_patients	2025-11-13T15:17:25.944770	rows=3
118	2	doctor	login	2025-11-13T15:17:25.886951	drbob logged in
117	1	admin	view_logs	2025-11-13T15:16:48.809108	count=116
116	1	admin	view_patients	2025-11-13T15:16:48.790189	rows=3
115	1	admin	login	2025-11-13T15:16:48.723640	admin logged in
114	1	admin	view_logs	2025-11-13T15:09:05.346157	count=113



3. CIA Triad Implementation

3.1 Confidentiality

- Sensitive data (name, contact) is either masked or encrypted.
- Role-based access controls ensure doctors see only anonymized data, receptionists have limited access, and admins have full access.

3.2 Integrity

- All modifications to patient data (add, edit, delete) are logged with user and timestamp.
- Audit trails allow detection of unauthorized changes.
- Anonymization ensures integrity of sensitive data while maintaining usability.

3.3 Availability

- System uptime is tracked.
- Regular database backups allow recovery in case of failure.
- Exception handling ensures the dashboard remains available to authorized users.

4. GDPR Alignment

- **Lawful Processing:** Data access is restricted to authorized roles.

- **Data Minimization:** Only necessary data is displayed per user role; sensitive information is anonymized.
- **Accountability:** Logs maintain full traceability of all actions.
- **Right to Privacy:** Masking and encryption protect patient confidentiality.
- **Data Security Measures:** Admins perform anonymization and backups to ensure compliance with GDPR requirements.

5. Conclusion

The hospital management system effectively integrates security and privacy measures using the CIA triad and aligns with GDPR principles. Role-based access, anonymization, logging, and backups ensure that patient data is confidential, integrity is maintained, and availability is guaranteed.

This report, combined with screenshots, demonstrates a fully implemented secure and privacy-compliant digital hospital system.

Demo Vedio Link:

https://drive.google.com/file/d/1pUB6FSznrBPvT_gv1DHKA1UQFJ48k1zP/view?usp=drive_link